



## **STUDYSYNC**

**A COLLABORATIVE GROUP STUDY PLATFORM**

### **SOFTWARE REQUIREMENTS SPECIFICATION & ANALYSIS**

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# 1. Introduction

The Collaborative Group Study Platform (CGSP) is envisioned as an all-in-one solution to address the evolving needs of modern study groups. With the shift towards virtual and hybrid learning environments, there is a growing demand for platforms that facilitate effective collaboration. CGSP combines advanced communication tools, resource sharing, and group management features to create a seamless collaborative experience.

## 1.1 Purpose

The Collaborative Group Study Platform (CGSP) aims to provide an intuitive and robust environment for students, educators, and professionals to collaborate effectively in study groups. The platform combines real-time communication, resource sharing, and group management to enhance the learning experience. By integrating exciting features like video conferencing and personalized group recommendations, CGSP facilitates collaborative learning in a seamless and efficient manner.

## 1.2 Intended Audience

The primary audience for this SRS document includes:

- **Developers:** To understand the functional and non-functional requirements for implementing the platform.
- **Project Managers:** To plan and allocate resources efficiently.
- **Stakeholders:** To review and provide feedback on the project's goals and requirements.
- **Quality Assurance Team:** To ensure all features are delivered as specified.
- **End Users:** Students, educators, and professionals who will use the platform for collaborative study purposes.

## 1.3 Usage Scenarios

### Scenario 1: Joining and Participating in a Group Study

1. A student logs into CGSP using their secure credentials.
2. The student browses recommended study groups based on their interests and joins a "Data Science" study group.

3. Inside the group dashboard, the student views the list of group members and recent activities.
4. The group admin assigns discussion topics, and members begin a real-time group chat to share ideas and resources.
5. One member shares a relevant PDF file, which is uploaded to the group's file repository for everyone to access.

### **Scenario 2: Hosting a Video Conference for Collaborative Learning**

1. An educator creates a group titled "Mathematics Revision" and invites students to join.
2. The educator schedules a video conference to explain complex topics with an interactive whiteboard.
3. During the session, attendees discuss the topics, share screens, and record the session for future reference.
4. Notifications are sent in real-time to remind members of the session.

### **Scenario 3: Personalized Group Recommendations and Notifications**

1. A professional enters their areas of interest (e.g., "Machine Learning") during profile setup.
2. CGSP suggests a list of active study groups matching their preferences.
3. Upon joining a group, the user receives notifications about new discussions, resource uploads, and upcoming video conferences.

## **1.4 Conclusion**

This document serves as a comprehensive reference for the requirements of CGSP. It ensures all stakeholders have a shared understanding of the project's objectives and scope, paving the way for successful development and deployment.

## 2. Inception

### 2.1 Understanding the Problem

Collaborative learning is crucial in today's educational and professional environments. Existing tools either lack specific features tailored for study groups or fail to integrate multiple functionalities seamlessly. The CGSP is designed to address the following challenges:

- Fragmentation of tools for communication, resource sharing, and group management.
- Lack of personalized recommendations for group formation.
- Limited interactivity in virtual collaborative environments.
- Inadequate notification systems leading to missed activities.

### 2.2 Icebreaking

The CGSP bridges the gap between traditional and digital collaborative methods by fostering a unified platform that:

- Encourages real-time interactions.
- Supports diverse user needs, from basic communication to advanced video conferencing.
- Promotes resource sharing and efficient group management.
- Provides timely notifications and personalized suggestions for a better user experience.

### 2.3 Identifying the Stakeholders

The key stakeholders for CGSP are:

- Student: As primary users, they rely on the platform for study group formation, discussions, and collaboration.
- Teacher: They utilize the platform to guide study groups and share resources.

### 2.4 Stakeholder Viewpoints

After talking with the stakeholders, we collected the following viewpoints of requirements from them.

### **Student**

Students are looking for a platform that helps them discover and join study groups focused on advanced subjects. They want to share detailed academic resources, engage in topic-based discussions, and collaborate on projects. A structured platform that supports deep learning, effective organization of study sessions, and role-based group management is essential. Personalized group recommendations based on interests, timely notifications for activities, and tools that aid in both coursework and research projects will enhance their learning experience and collaboration opportunities.

### **Teacher**

Teachers seek a platform to create and moderate academic groups, guide students, and share educational materials. They want tools that help them manage group discussions, monitor activity, and encourage student participation. They may also value video conferencing for virtual class discussions or office hours.

## **2.5 Working Towards Collaboration**

Collaborative efforts are focused on:

- Aligning the platform's features with stakeholder needs.
- Establishing open channels for feedback throughout the development lifecycle.
- Encouraging active involvement of key stakeholders during testing phases to refine the platform's usability and functionality.

## **2.6 Elicitation of CGSP (Collaborative Group Study Platform)**

The following CGSP were identified during requirements gathering:

- Real-time Communication: Support for group chat and video conferencing.
- Resource Sharing: Secure upload, storage, and sharing of files.
- Group Management: Roles, permissions, and dashboards for effective group interaction.
- Notifications: Real-time and activity-specific alerts.
- Personalization: Recommendations for group formation based on user preferences.

### 3. Quality Function Deployment

Quality Function Deployment (QFD) is a systematic approach to translate the needs of users into measurable technical requirements. By employing QFD, the platform ensures that customer expectations are transformed into design specifications that align with the goals of CGSP. This methodology helps maximize user satisfaction by bridging subjective user needs with objective, quantifiable criteria. The requirements listed above were identified and refined successfully using QFD to meet the collaborative and technical demands of the platform.

#### Collaborative Requirements Gathering

Requirements for CGSP were gathered through collaborative workshops involving representatives from all stakeholder groups. This approach ensured:

- A holistic understanding of user needs.
- Prioritization of features based on stakeholder input.
- Early identification of potential challenges and mitigation strategies

### 3.1 Normal Requirements

#### User Authentication & Management:

- **Sign Up/Log In:** Users can create accounts using email, password etc.
- **User Profiles:** Users can see and edit their profiles and information.
- **Secure Authentication:** JWT-based authentication for secure user sessions.

#### Group Management:

- **Create/Join Study Groups:** Users can create a new group or join groups using invitation.
- **Group Dashboard:** Display group details, member list, and recent activity in a basic layout. Admin only layout for group update.
- **Role Management:** Allow group creators to assign admin roles with limited options (e.g., add members or remove members).

#### Real-time Communication:

- **Group Chat:** Basic real-time text-based chat using sockets.
- **Post Discussions:** Users within a group can post discussions on certain topics.

**Resource Sharing:**

- **Upload and Share Files:** Basic file upload feature (e.g., PDFs, images) within groups.
- **File Repository:** A central place to store and access shared resources like documents and study guides.

**Notification System:**

- **Real-time Notifications:** Users receive real-time notifications for important activities, such as new members added in group, and new discussion posts.
- **Activity Alerts:** Notify users when they are added to a new group, assigned a role, or removed from a group.

## 3.2 Exciting Requirements

- **Video Conferencing:** Users can host real-time video calls with screen sharing, interactive whiteboards, and recording features.
- **Personalized Group Recommendation:** Allow users to input specific subjects or topics of interest, which are then used to recommend personalized group suggestions.

## 4. User Story

### User Management

- **Authentication** : Users can create their profiles by providing their name, email address, and password. The system stores the users credentials in the database, completing the registration process. Users can log into their accounts afterward using their email and password, with the system authenticating these credentials against the database. If a user forgets their password, they can use the "Forgot Password?" feature, which guides them through a password reset process. Users also have the option to change their password from their account settings. They can log out of their account anytime they want to end their session.
- **Dashboard** : Users can view their dashboard with detailed information about their profile. They can add images of certificates, grades, and resumes or CVs to the dashboard. Users have the option to hide specific details on their dashboard, ensuring that other users cannot view that information.

### Group Management

- **Group Creation** : Any registered student/teacher can create a study group by providing a group name, specifying the subject or topic focus, and selecting the group status (public or private). The creator, who assumes the role of the group admin, can invite other users to join the group by sending an invitation using their user ID. The creator also has the option to make the group private, which means the group will not be visible in the search results.
- **Group Join** : A user can join a group by searching for it using the search panel. After the user clicks the "Join" button, a notification will be sent to the admin of that group. If the admin accepts the request, the user will be able to join the group.
- **Group Admin** : The student who creates the group automatically becomes an admin. Group admins can assign other members as

additional admins to help manage the group. They have the ability to add new members, remove inactive ones, and update group details, such as the description and focus area.

## Communication

Members of a group can interact using a real-time chat system, allowing instant communication. The chat system supports text messaging, with users able to share URLs and resource links directly in the chat. Group admins can post discussion topics to discuss specific matters. Admin needs to close that discussion. Discussions are visible in the chat panel for their duration. Group members can search for a discussion by topic name and will be placed at the start of that discussion in the chat panel.

## Resource Sharing

- **File Uploading:** Members can upload study materials such as PDFs, images, notes, or external resource links directly into the group.
- **File Repository:** Uploaded files are stored in a central repository for each user. Files are organized within the repository. Users can download or delete files from their repository. Each file contains the metadata of file name, file type, file size, file description, upload time.

## Notification

Users receive real-time notifications for the following events:

- A new member joins or creates a group
- A new discussion thread is started
- A file is shared
- Group join requests (for admins)
- A group starts video conferencing

Users also receive invitations via notifications, which they can accept or reject. Each notification contains its sender (if sent by someone), notification content.

### **Video Conferencing ( Exciting )**

Group members can initiate video conferences directly within the group panel. The system supports basic video and audio calls, allowing participants to communicate in real-time. Members receive notifications when a video call starts, inviting them to join.

### **Group Recommendation and Search**

Users can discover groups through recommendations on the dashboard or by searching in the group search panel. Users can filter groups by topics, size, majority members.

## **5. Scenario Based Modeling**

### **5.1 Use Case Diagram**

A Use Case describes the system behavior under various conditions as the system responds to a request from one of its stakeholders. In fact, a use

case diagram is a kind of visualization of the system where an end-user has an idea of a specific feature. It simply describes a story

using corresponding actors who perform important roles in the story and makes the story understandable for the users.

The first step in writing a Use Case is to define that set of “actors” that will be involved in the story. Actors are the different people or systems that use the system or product within the context of the function and behavior that is to be described. Actors represent the roles that people play as system operators. They produce some information or consume some information. Every user has one or more goals when using the system.

**Primary Actor:** Primary actors interact directly to achieve the required system function and derive the intended benefit from the system. They work directly with the software. They produce some information and consume some information too.

**Secondary Actor:** Secondary actors support the system so that primary actors can do their work. They either produce or consume information. Here is given the use case diagram to observe the non-technical view of the system.

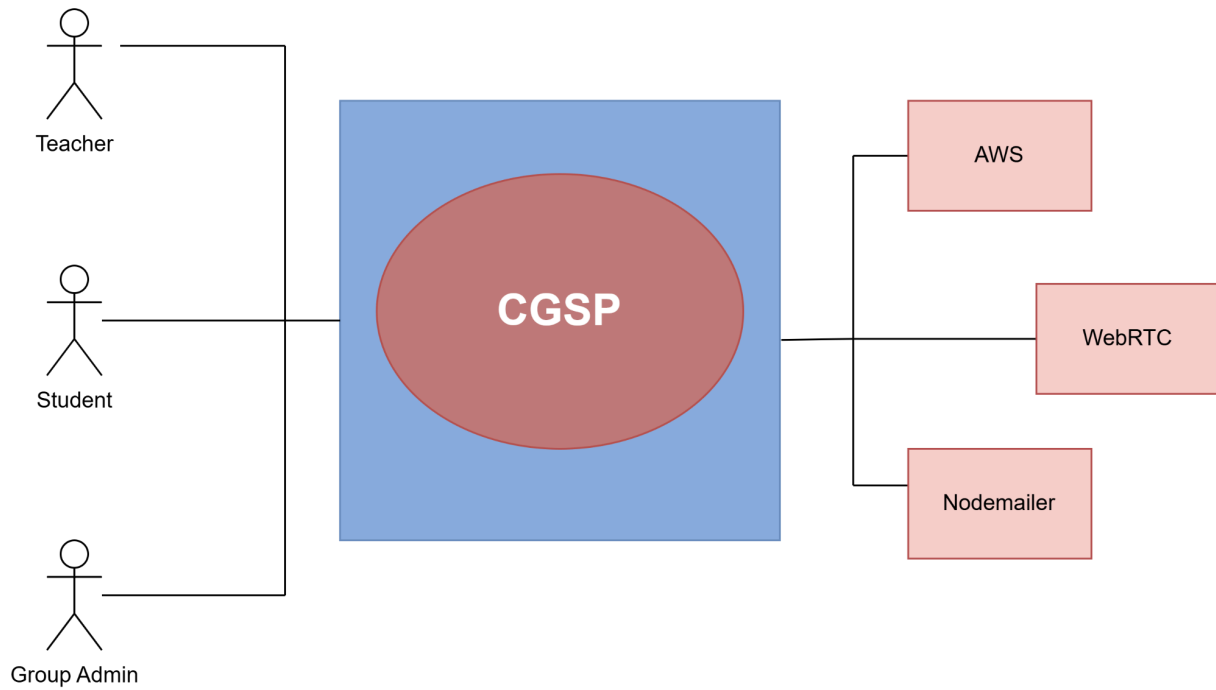
**Level: 0**

**USE CASE ID: 0**

**Name:** CGSP

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** AWS, Nodemailer, WebRTC



**Fig 1:** Collaborative Group Study Platform

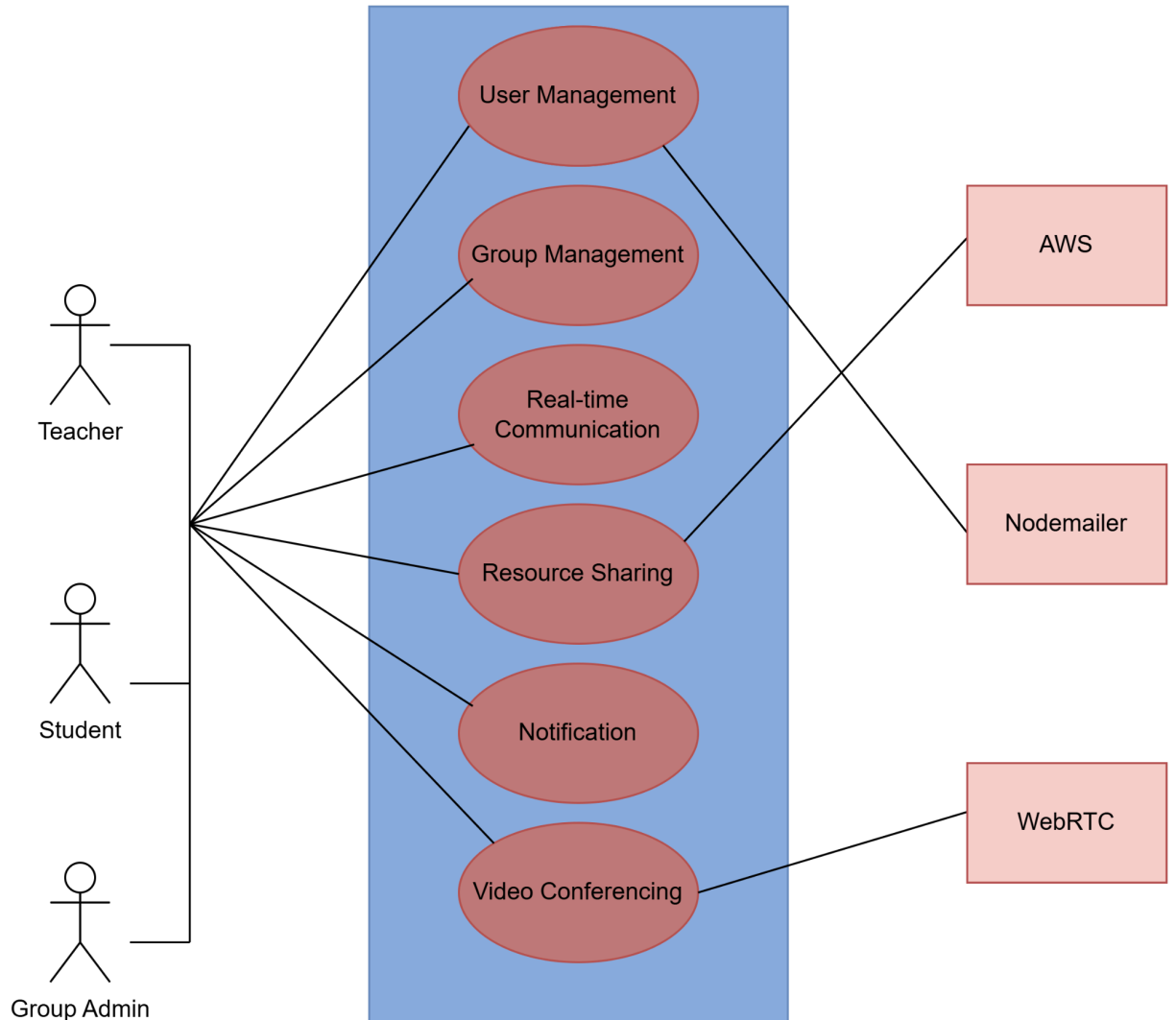
**Level: 1**

**USE CASE ID: 1**

**Name:** CGSP

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** AWS, Nodemailer, WebRTC



**Fig 2:** Level 1 of Collaborative Group Study Platform

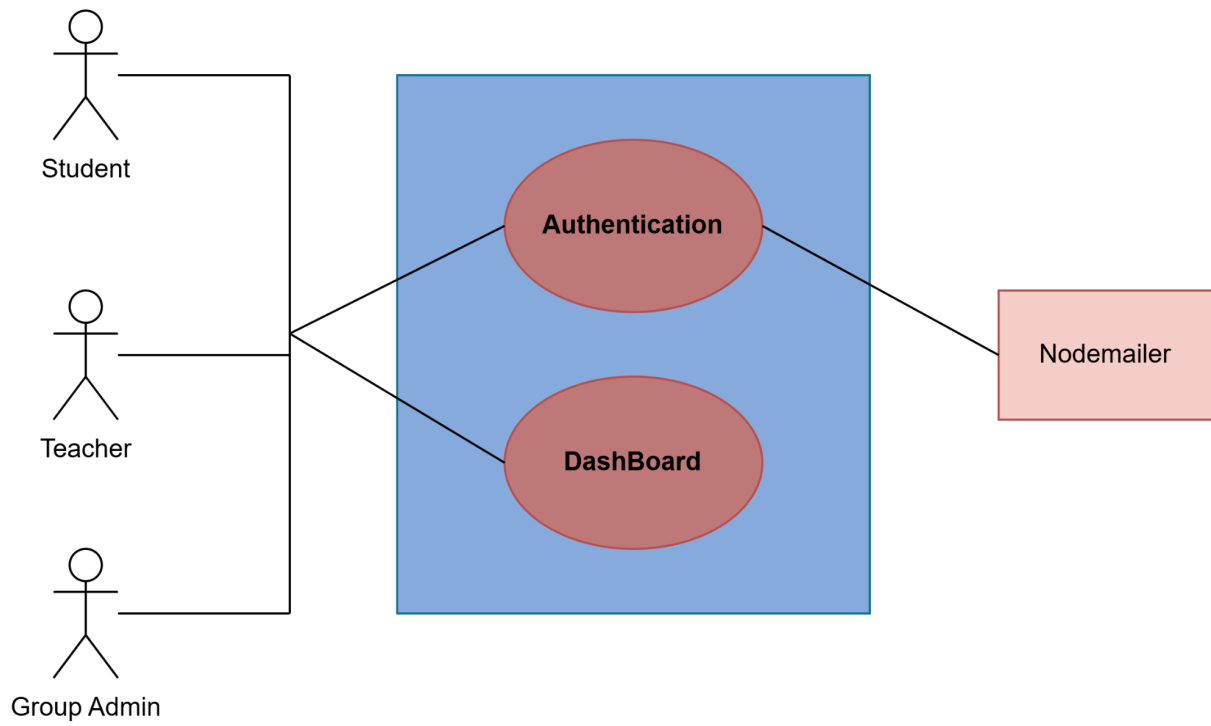
**Level: 1.1**

**USE CASE ID: 1.1**

**Name:** User Management

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** None



**Fig 3:** User Management

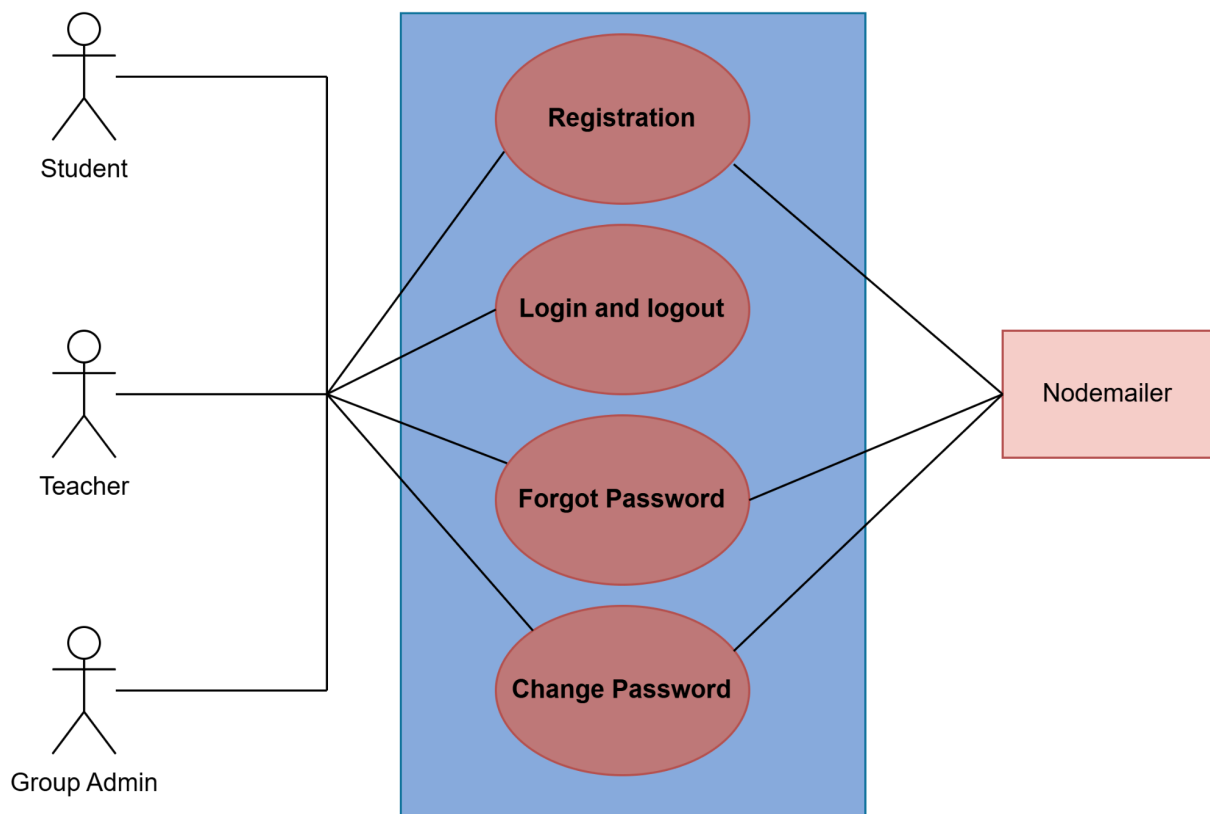
**Level: 1.1.1**

**USE CASE ID: 1.1.1**

**Name:** Authentication

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** None



**Fig 4: Authentication**

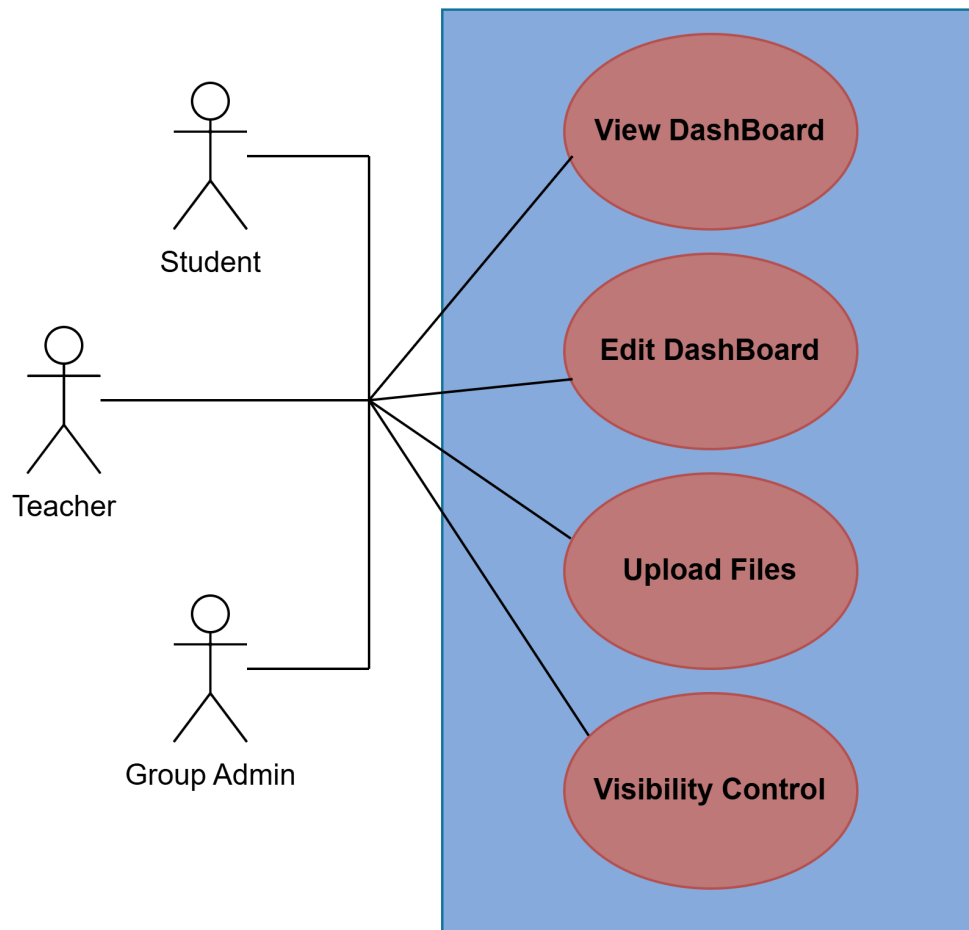
**Level: 1.1.2**

**USE CASE ID: 1.1.2**

**Name:** DashBoard

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** None



**Fig 5:** DashBoard

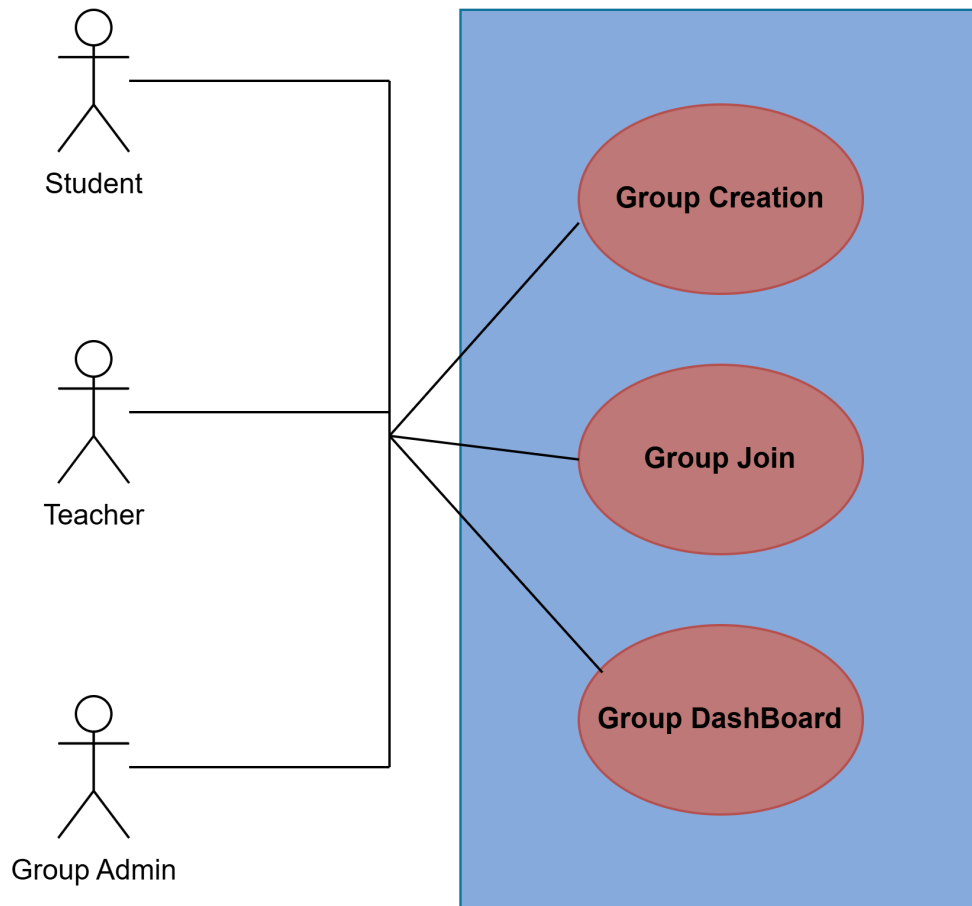
**Level:** 1.2

**USE CASE ID:** 1.2

**Name:** Group Management

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** None



**Fig 6:** Group Management

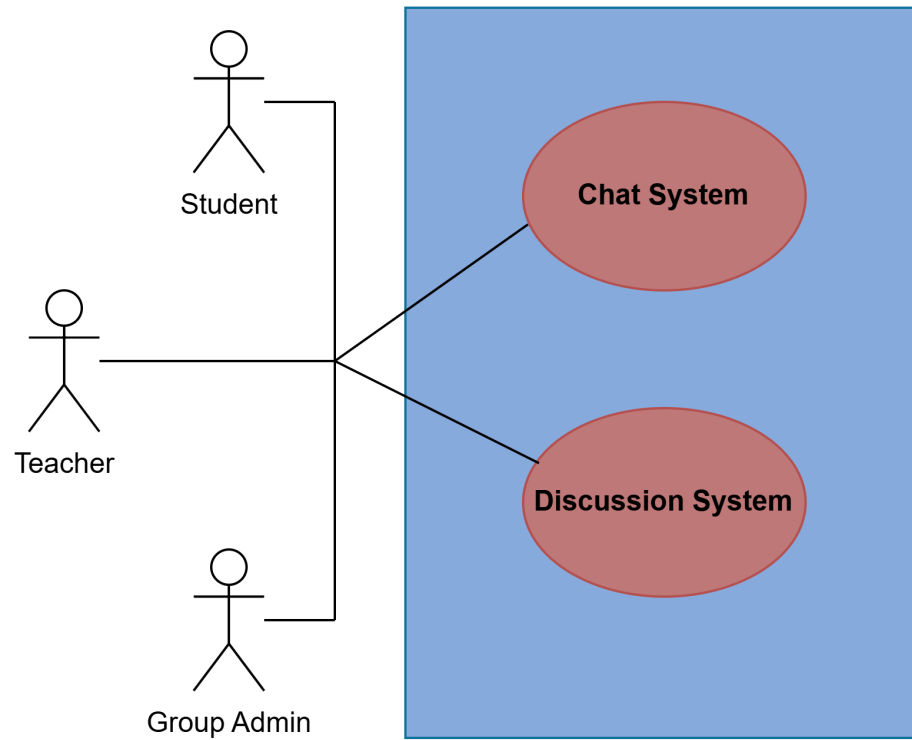
**Level:** 1.3

**USE CASE ID:** 1.3

**Name:** Real-time Communication

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** None



**Fig 7:** Real-time Communication

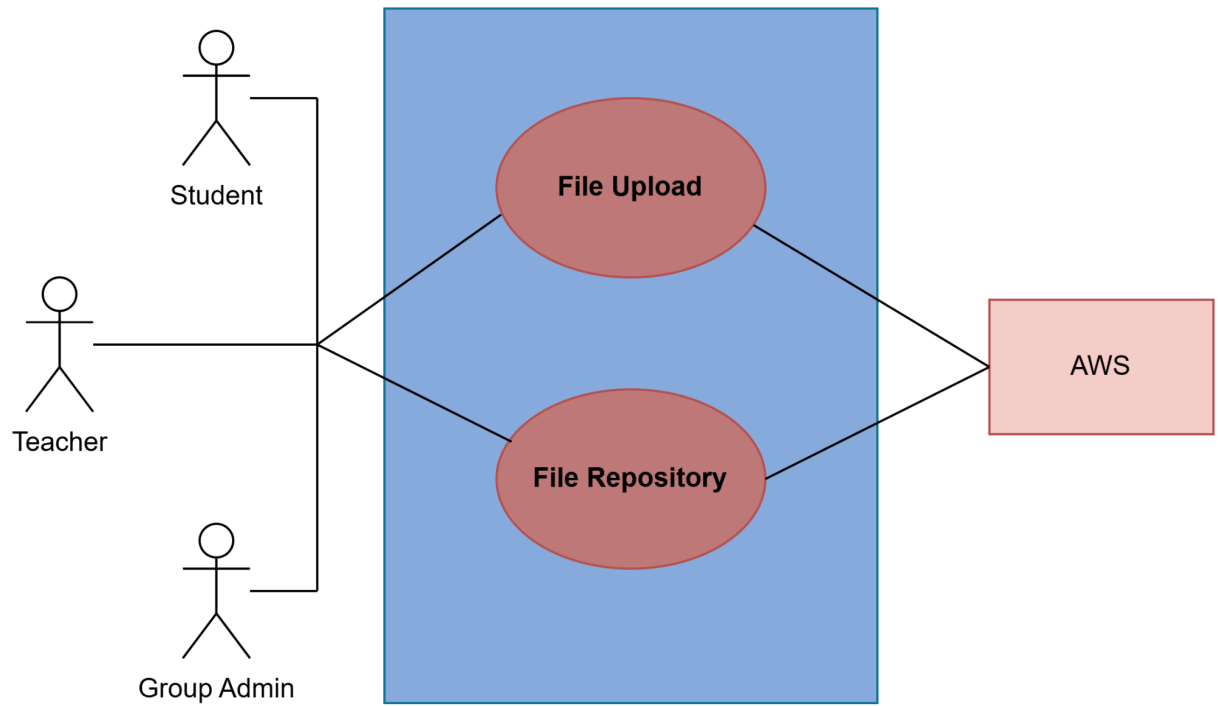
**Level:** 1.4

**USE CASE ID:** 1.4

**Name:** Resource Sharing

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** AWS



**Fig 8:** Resource Sharing

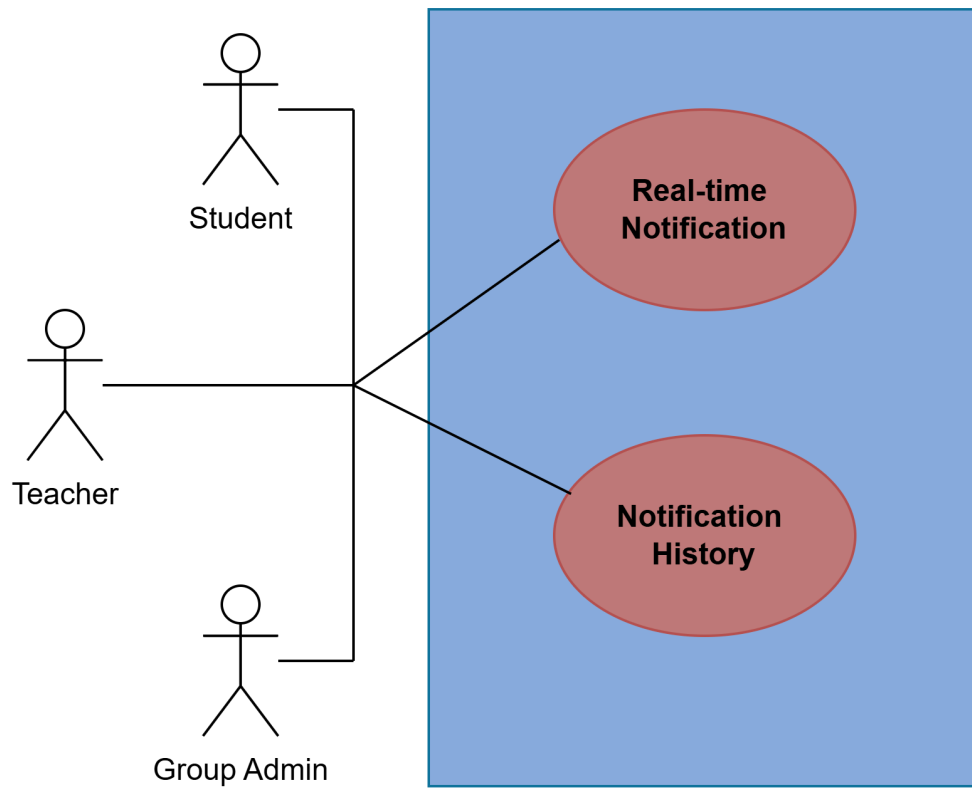
**Level:** 1.5

**USE CASE ID:** 1.5

**Name:** Notification System

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** None



**Fig 9:** Notification System

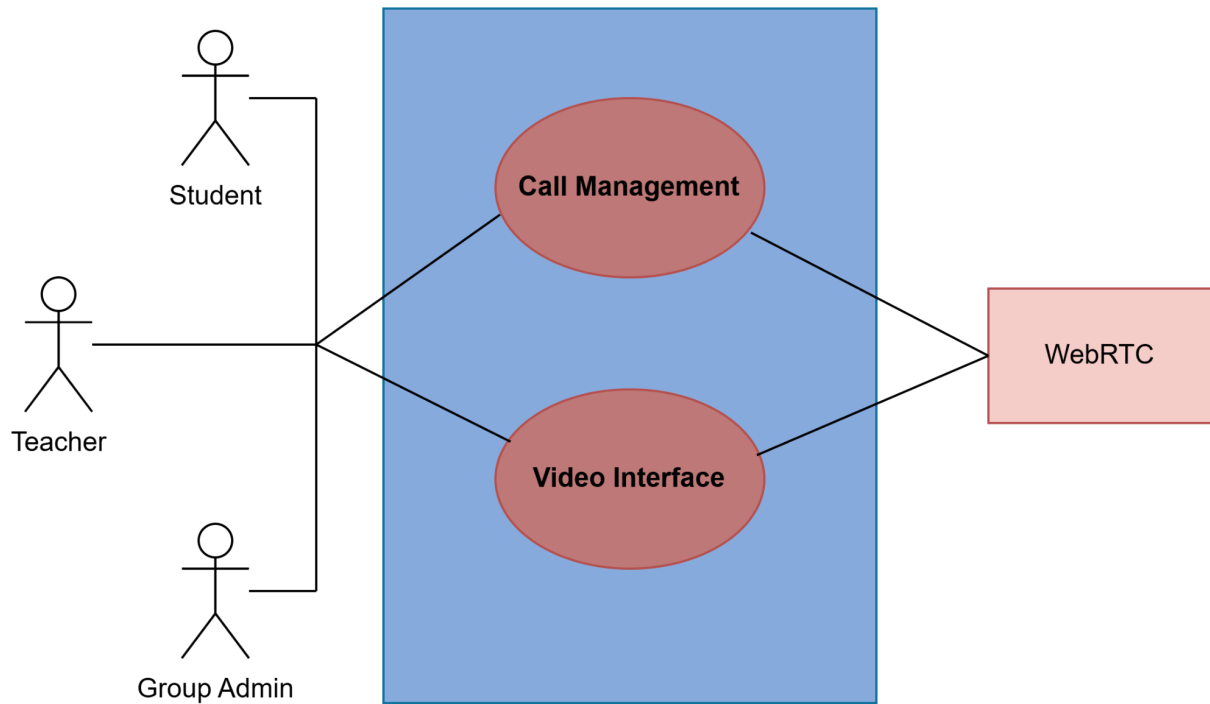
**Level:** 1.6

**USE CASE ID:** 1.6

**Name:** Video Conferencing

**Primary Actor:** Teacher, Student, Group Admin

**Secondary Actor:** WebRTC



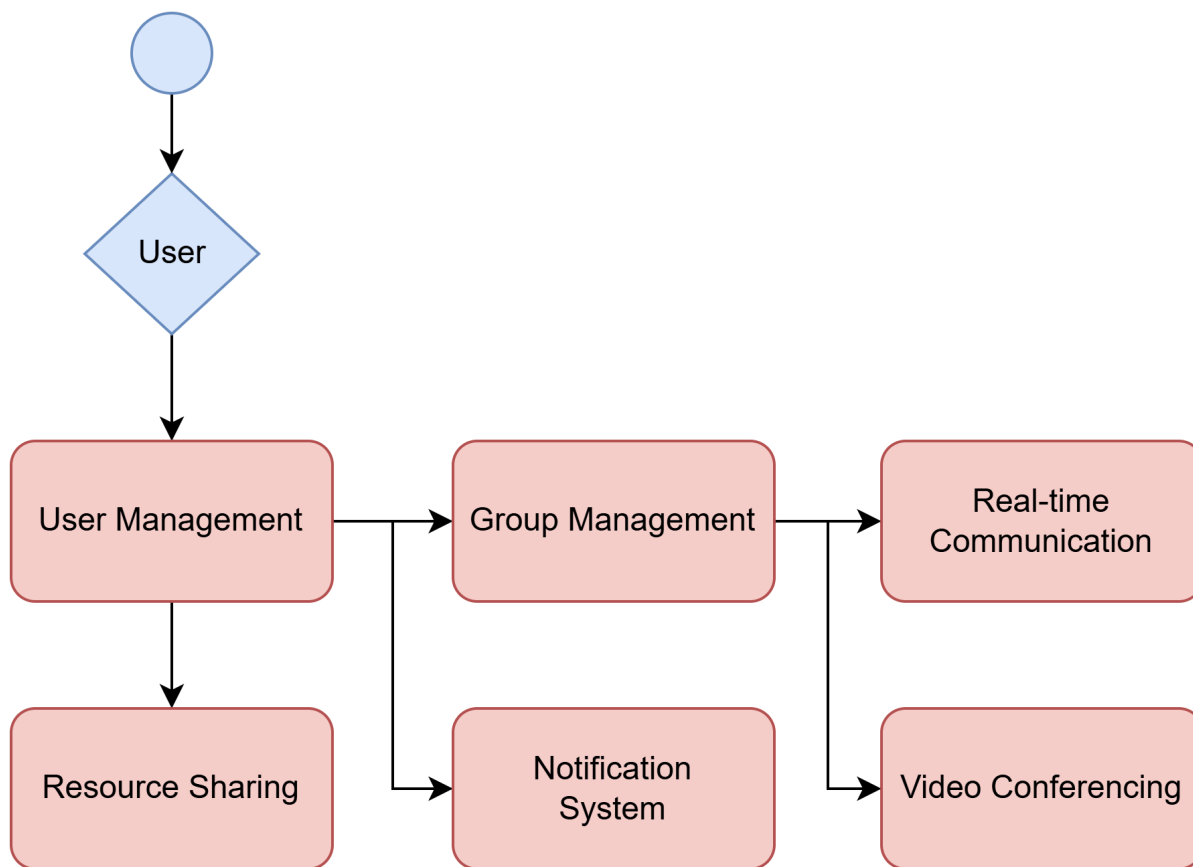
**Fig 10:** Video Conferencing

## 5.2 Activity Diagram

**Level: 1**

**Name:** CGSP

**Reference:** Use case Diagram level-1

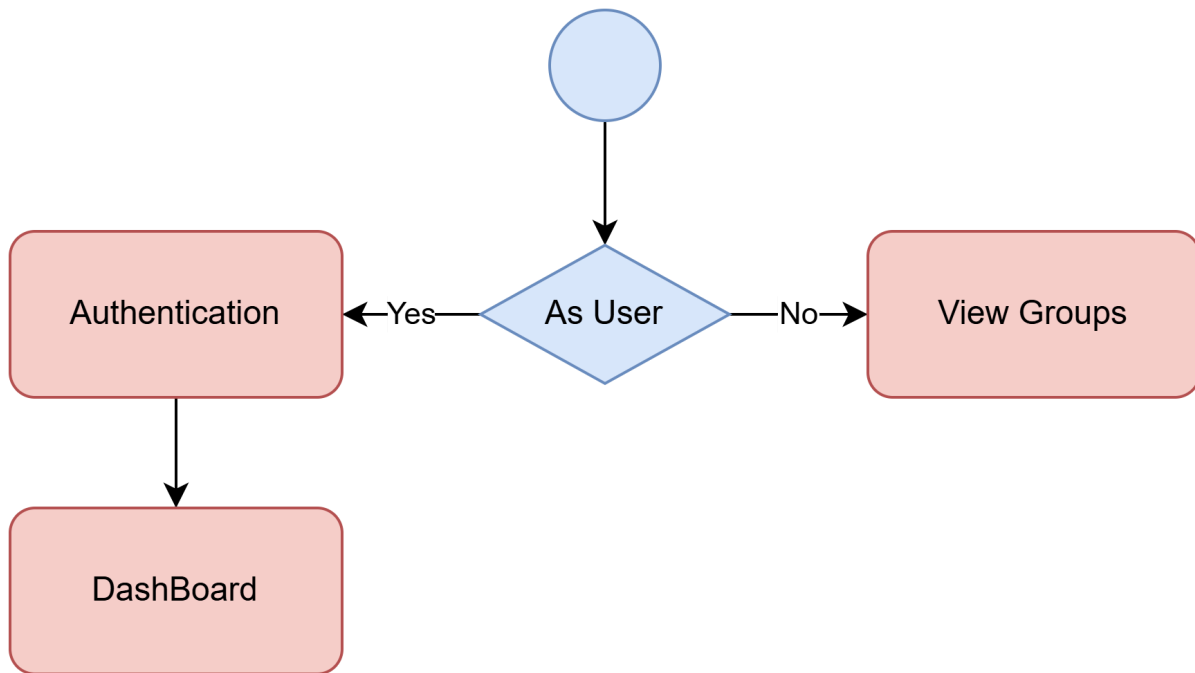


**Fig 11:** Collaborative Group Study Platform (Use Case - 1)

**Level:** 1.1

**Name:** User Management

**Reference:** Use case Diagram level-1.1

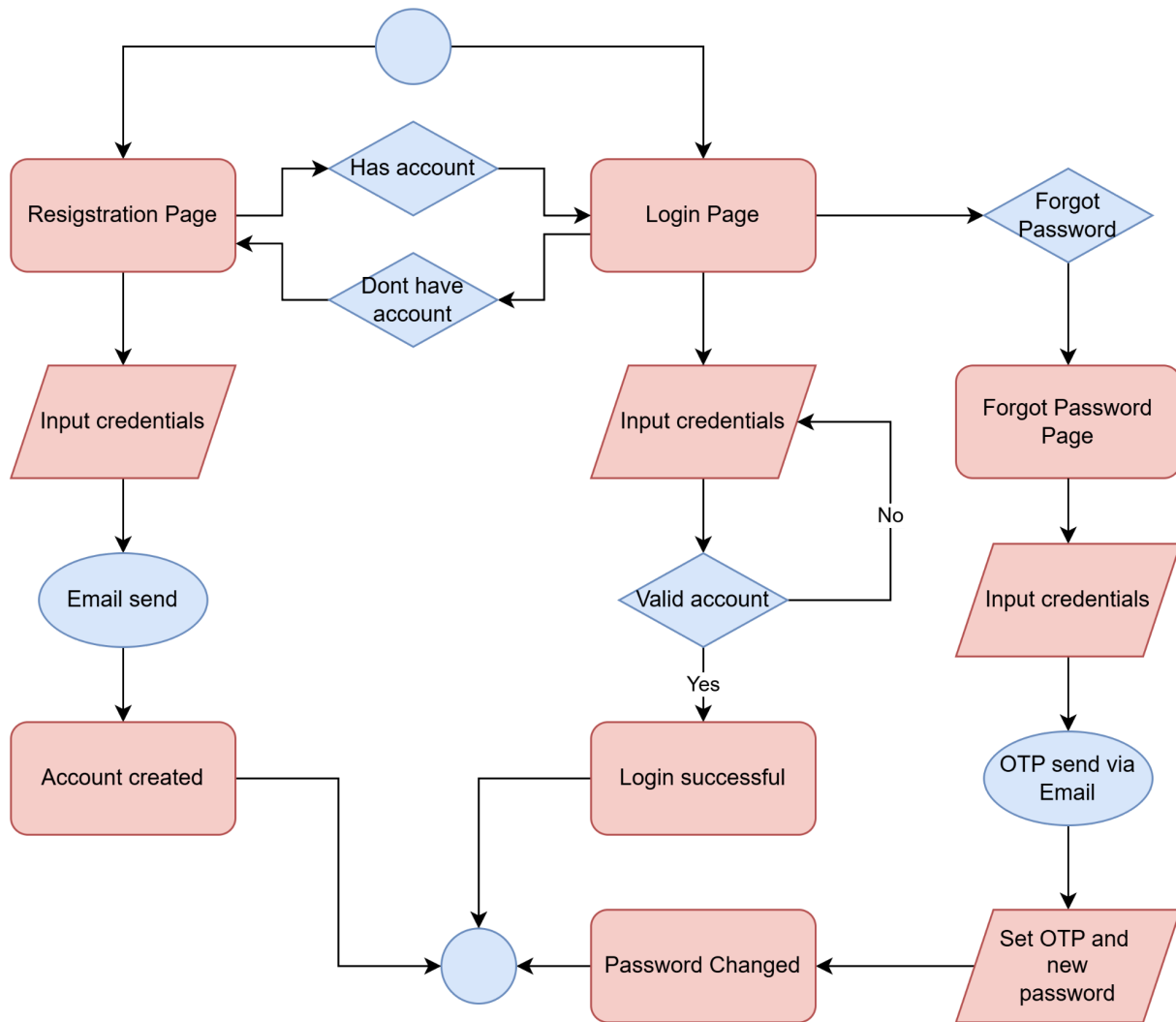


**Fig 12:** User Management (Use Case - 1.1)

**Level:** 1.1.1

**Name:** Authentication

**Reference:** Use case Diagram level-1.1.1

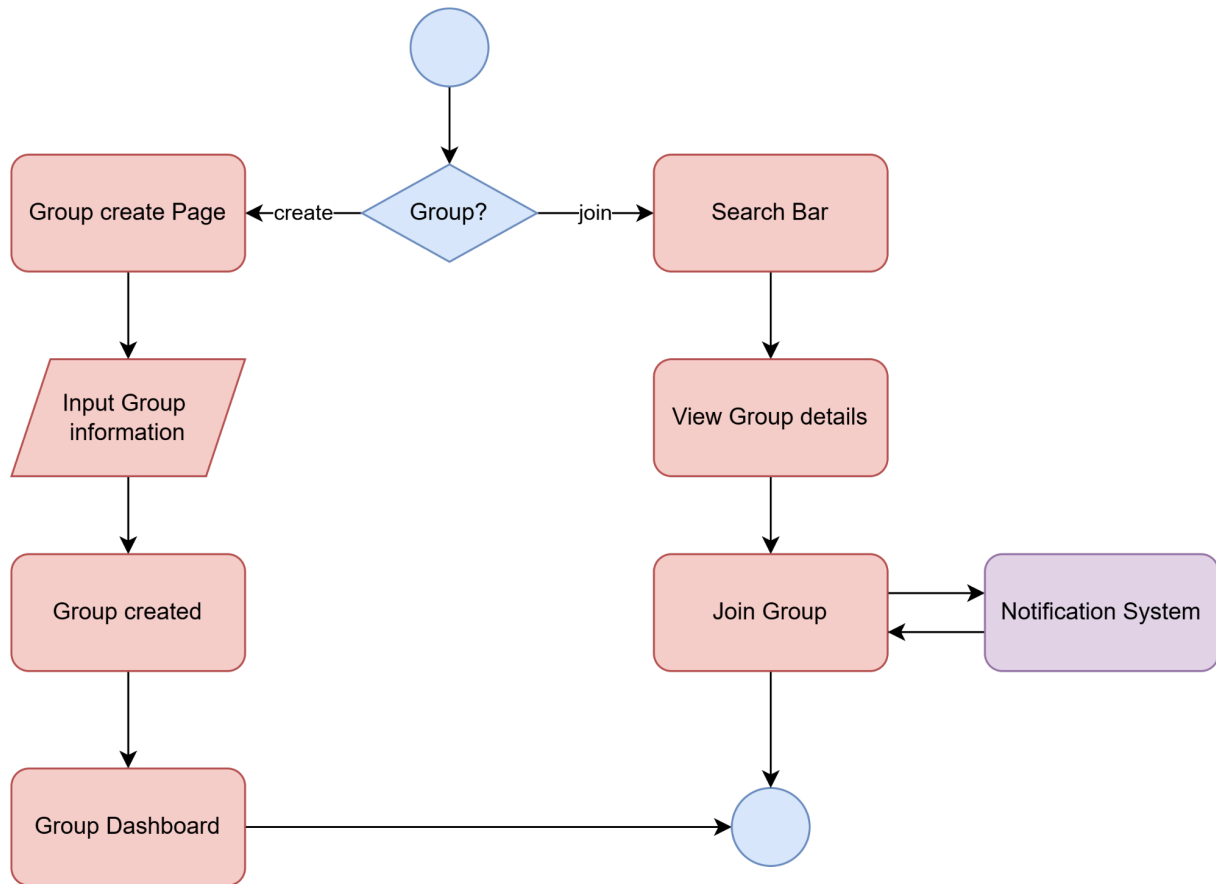


**Fig 13:** Authentication(Use Case - 1.1.1)

**Level: 1.2**

**Name:** Group Management

**Reference:** Use case Diagram level-1.2

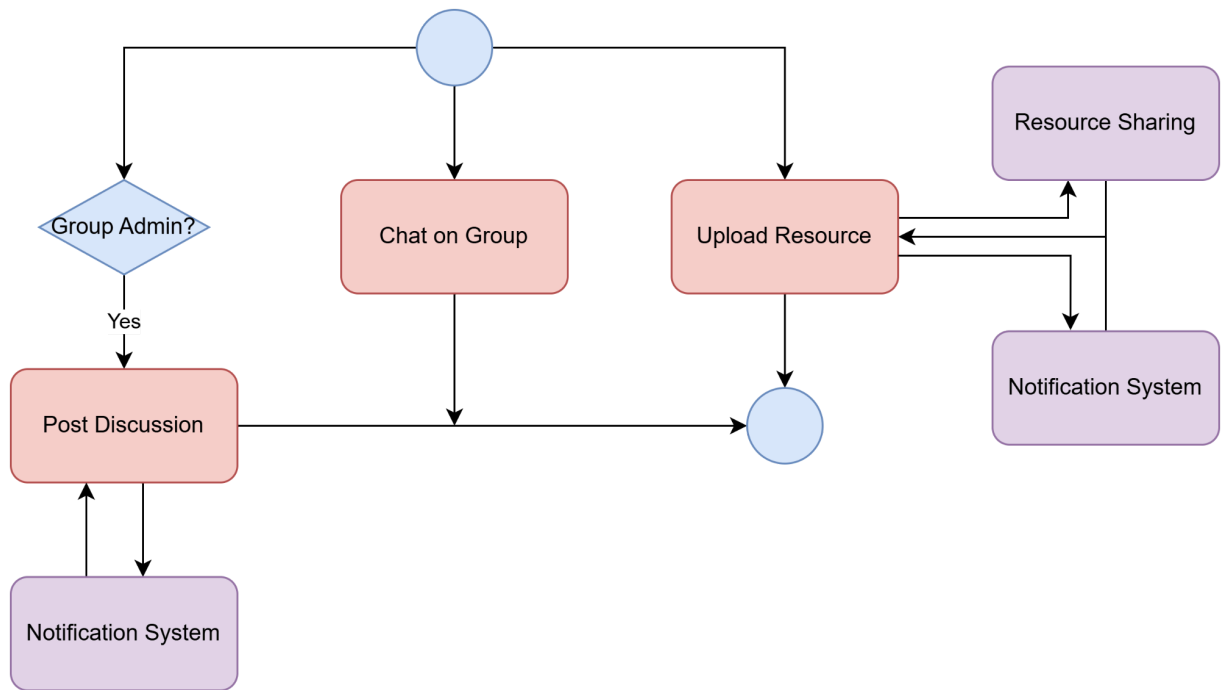


**Fig 14:** Group Management (Use Case - 1.2)

### Level: 1.3

**Name:** Real-time Communication

**Reference:** Use case Diagram level-1.3

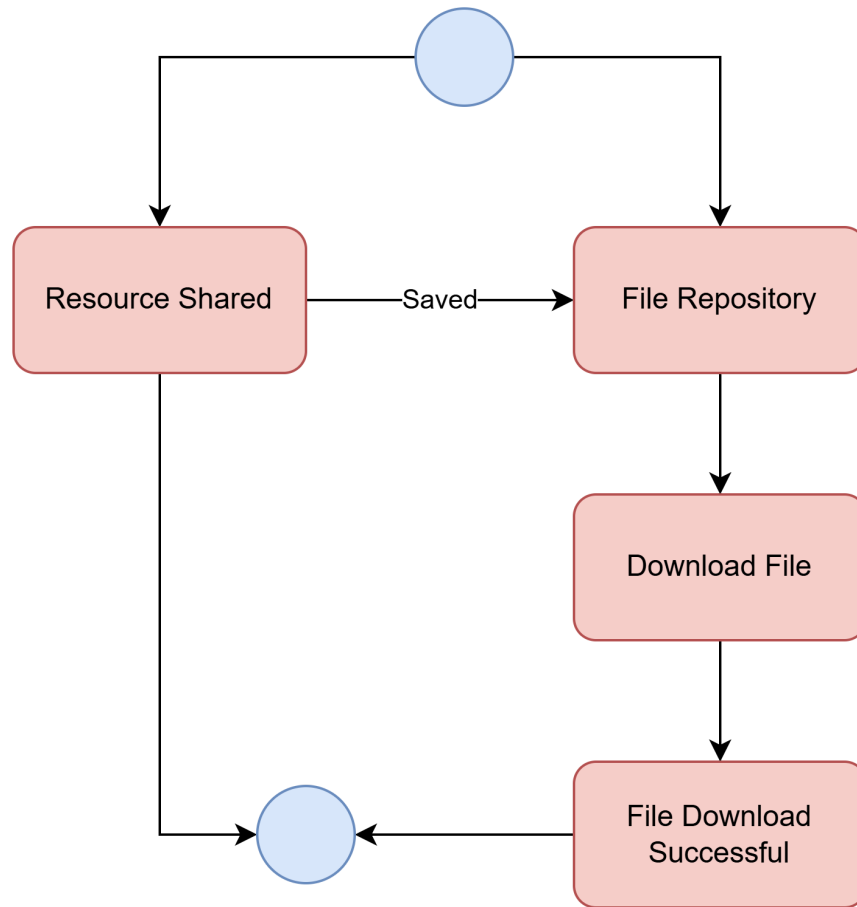


**Fig 15:** Real-time Communication (Use Case - 1.3)

**Level: 1.4**

**Name:** Resource Sharing

**Reference:** Use case Diagram level-1.4

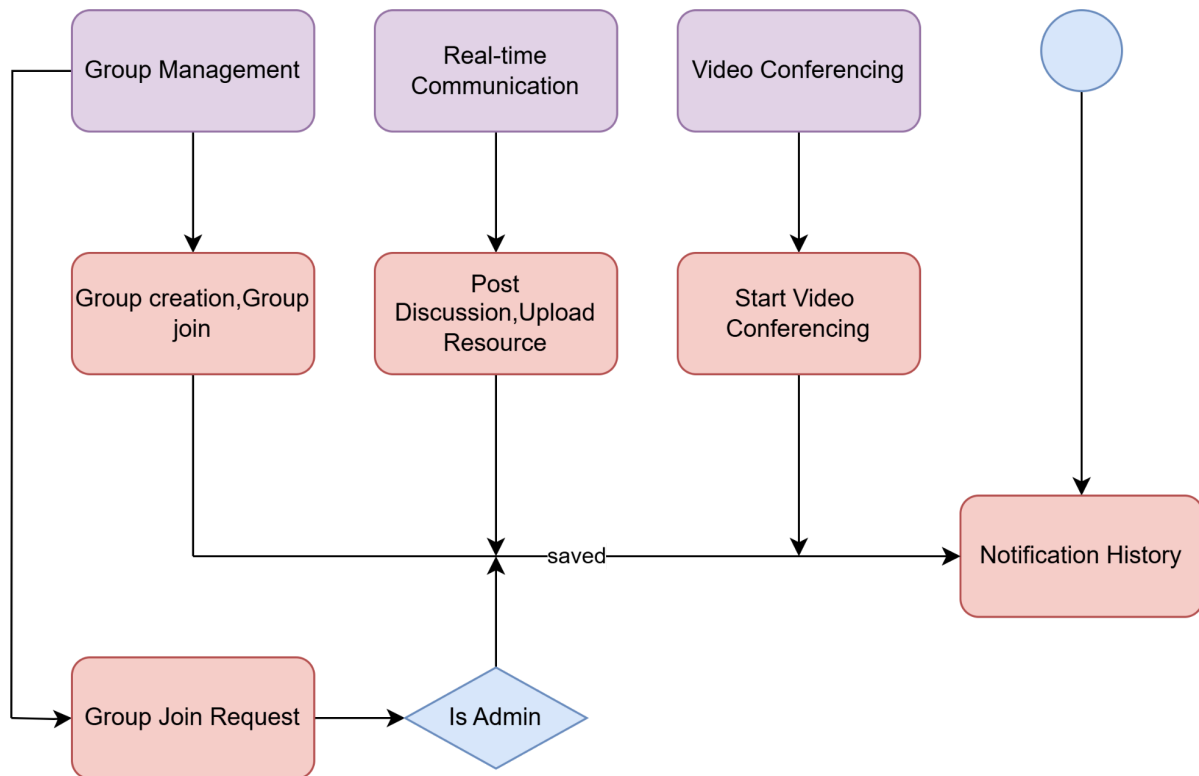


**Fig 16:** Resource Sharing (Use Case - 1.4)

**Level: 1.5**

**Name:** Notification System

**Reference:** Use case Diagram level-1.5

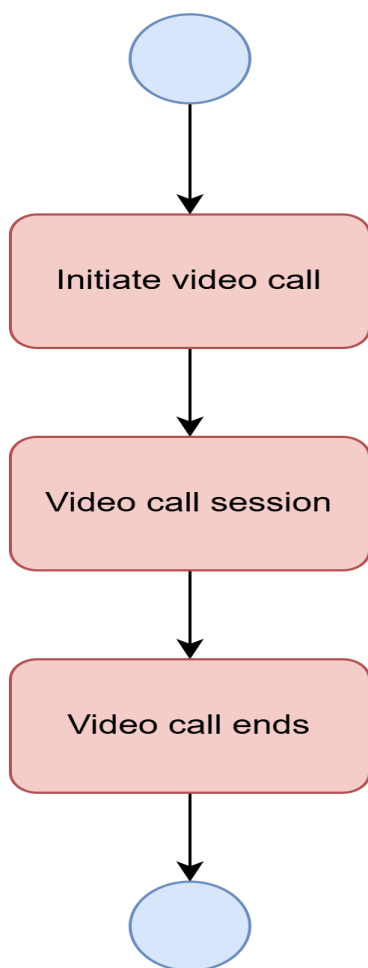


**Fig 17: Notification System (Use Case - 1.5)**

**Level: 1.6**

**Name:** Video Conferencing

**Reference:** Use case Diagram level-1.6



**Fig 18:** Video Conferencing (Use Case - 1.6)

## 6. Data Based Modeling

**DATA MODELING CONCEPT** : If software requirements include the necessity to create, extend or interact with a database or complex data structures need to be constructed and manipulated, then the software team chooses to create data models as part of overall requirements modeling. The entity relationship diagram (ERD) defines all data objects that are processed within the system, the relationships between the data objects and the information about how the data objects are entered, stored, transformed and produced within the system.

**DATA OBJECTS** : A data object is a representation of composite information that must be understood by the software. A data object can be an external entity, a thing, an occurrence, a role, an organizational unit, a place or a structure.

### 6.1 Noun Identification:

| Serial | Nouns          | Problem /Solution space | Attributes                     |
|--------|----------------|-------------------------|--------------------------------|
| 1      | User           | S                       | 4, 5, 6,13, 23, 27, 31         |
| 2      | Authentication | S                       |                                |
| 3      | Profile        | S                       | 14, 16, 17, 18, 19, 20, 27, 63 |
| 4      | Name           | S                       |                                |
| 5      | Email          | S                       |                                |
| 6      | Password       | S                       |                                |
| 7      | System         | P                       |                                |
| 8      | Credential     | P                       |                                |

|    |                         |   |                                        |
|----|-------------------------|---|----------------------------------------|
| 9  | Database                | P |                                        |
| 10 | Dashboard               | P |                                        |
| 11 | Account                 | P |                                        |
| 12 | Account Settings        | P |                                        |
| 13 | Session                 | S |                                        |
| 14 | Detailed information    | S |                                        |
| 15 | Logout                  | P |                                        |
| 16 | Images of certificate   | P |                                        |
| 17 | Images of grades        | S |                                        |
| 18 | Images of resume        | S |                                        |
| 19 | Images of CV            | P |                                        |
| 20 | Option ( Hide details ) | P |                                        |
| 21 | Information             | P |                                        |
| 22 | Student                 | P |                                        |
| 23 | Study group             | S | 22, 24, 25, 26, 27, 32, 33, 34, 35, 37 |
| 24 | Group name              | S |                                        |
| 25 | Subject or topic focus  | S |                                        |
| 26 | Group status            | S |                                        |

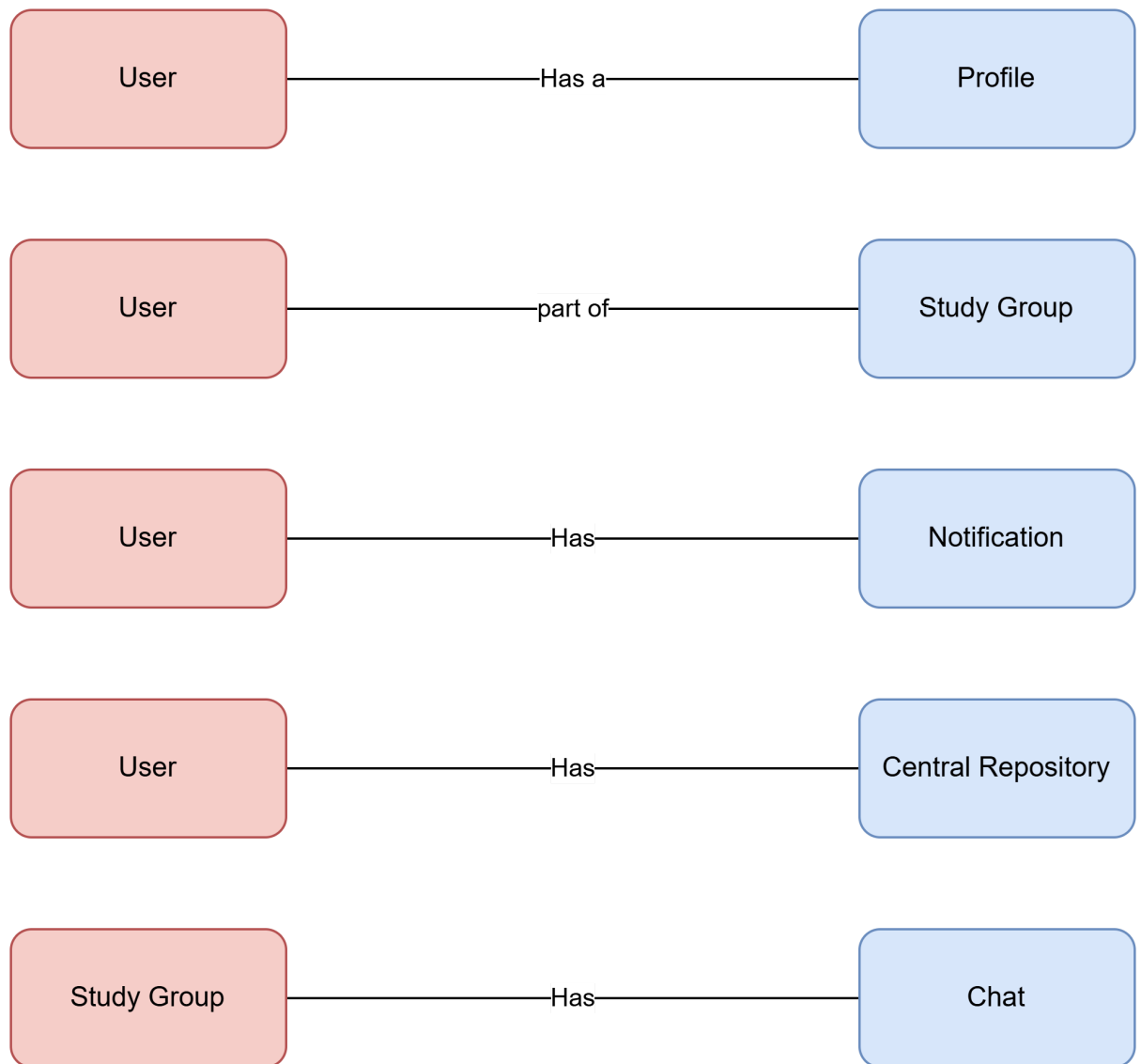
|    |                    |   |                                |
|----|--------------------|---|--------------------------------|
| 27 | User id            | S |                                |
| 28 | Invitation         | S |                                |
| 29 | Search result      | P |                                |
| 30 | Search panel       | P |                                |
| 31 | Notification       | S | 28, 41, 55, 56, 57             |
| 32 | Group admin        | P |                                |
| 33 | Member             | S | 27, 63                         |
| 34 | Group details      | P |                                |
| 35 | Description        | S |                                |
| 36 | Communication      | S |                                |
| 37 | Chat-system        | S | 38, 39, 40, 41, 43, 44, 46, 55 |
| 38 | Text message       | S |                                |
| 39 | URL                | S |                                |
| 40 | Resource-link      | S |                                |
| 41 | Discussion topic   | S |                                |
| 42 | Chat-panel         | S |                                |
| 43 | Study materials    | P |                                |
| 44 | Files              | S | 58, 59, 60, 61, 62             |
| 45 | Central repository | S | 43, 44                         |
| 46 | Discussion thread  | P |                                |

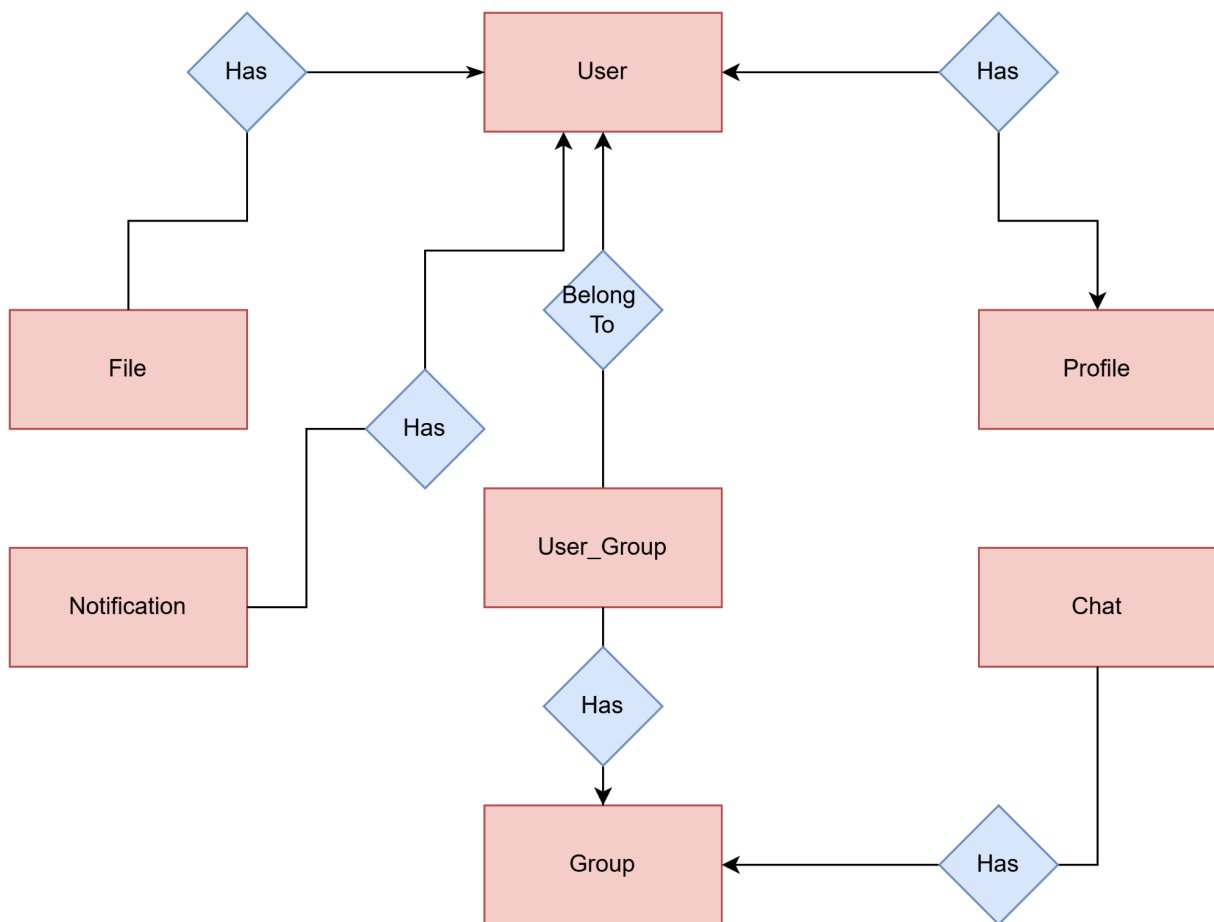
|    |                              |   |  |
|----|------------------------------|---|--|
| 47 | Group panel                  | S |  |
| 48 | Video call                   | S |  |
| 49 | Audio call                   | S |  |
| 50 | Recommendation               | S |  |
| 51 | Group search                 | S |  |
| 52 | Topics (filter)              | S |  |
| 53 | Size (filter)                | S |  |
| 54 | Majority members<br>(filter) | S |  |
| 55 | Content                      | S |  |
| 56 | Group join                   | S |  |
| 57 | Group join request           | S |  |
| 58 | File type                    | S |  |
| 59 | File size                    | S |  |
| 60 | File name                    | S |  |
| 61 | File description             | S |  |
| 62 | Upload time                  | S |  |
| 63 | Role                         | S |  |

## 6.2 Final Data Objects:

1. User
2. Profile
3. Group
4. Notification
5. Chat
6. File

## 6.3 Relations:



**Fig 19: Entity Relation****6.4 ER DIAGRAM:****Fig 20: ER Diagram****6.5 Schema Diagram:**

| Data Object | Attribute                                                                                                                                                                                                                                                                                                                                                                       | Type                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User        | <u>user_id</u><br>email<br>password<br>name<br>session_token<br>status<br>last_login<br>reset_token<br>reset_token_expiry<br>created_at                                                                                                                                                                                                                                         | string<br>string<br>string<br>string<br>string<br>string<br>date<br>string<br>date<br>date                                                                                                                                                   |
| Profile     | <u>user_id</u><br>date_of_birth<br>gender<br>phone_number<br>profile_picture<br>address<br>bio<br>hobby<br>education<br>role<br>is_visible<br><br>If role = 'student'<br>– course_of_study<br>– current_year_or_semester<br>– department<br>– skills<br>– grade_sheet<br>– resume<br><br>If role = 'teacher'<br>– subjects<br>– designation<br>– experience<br>– qualifications | objectId<br>date<br>string<br>string<br>string(URL)<br>string<br>string<br>string<br>string<br>string<br>string<br>boolean<br><br>string<br>string<br>string<br>[string]<br>string<br>string<br><br>[string]<br>string<br>string<br>[string] |
| Group       | <u>group_id</u><br>group_name<br>group_description<br>group_size<br>group_status<br>type                                                                                                                                                                                                                                                                                        | string<br>string<br>string<br>number<br>string<br>string                                                                                                                                                                                     |

|              |                                                                                                                                                                                                                                         |                                                                                                |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
|              | topics<br>group_image<br>created_by<br>created_at                                                                                                                                                                                       | [string]<br>string<br>string<br>date                                                           |
| User_Group   | <u>group_id</u><br><u>user_id</u><br>role<br>joined_at                                                                                                                                                                                  | string<br>string<br>string<br>date                                                             |
| Notification | <u>user_id</u><br><u>notification_id</u><br>type(invitation,discussion_topic,<br>join_request,join_group,<br>file_shared,<br>video_conferencing)<br>content<br><u>sender</u> (user_id)<br><u>group_id</u><br>receive_date               | string<br>string<br>string<br><br>string<br>string<br>string<br>date                           |
| Chat         | <u>chat_id</u><br>type('text_message', 'link',<br>'discussion_topic', 'files',<br>'video_conferencing', 'join_group'<br>)<br>topic<br>discussionStatus<br>content<br><u>sender</u> (user_id)<br>file(URL)<br><u>group_id</u><br>send_at | string<br>string<br><br>string<br>string<br>string<br>string<br>string<br>date                 |
| File         | <u>file_id</u><br><u>user_id</u><br><u>group_id</u><br>file_name<br>file_type<br>file_size<br>file_path (URL)<br>uploaded_at<br><u>uploaded_by</u><br>description                                                                       | string<br>string<br>string<br>string<br>string<br>number<br>string<br>date<br>string<br>string |

## 7. Class Based Diagram

### Class-Based Modeling Concepts:

Class-based modeling represents the objects that the system will manipulate, the operations that will be applied to the objects, relationships between the objects and the collaborations that occur between the classes that are defined.

#### 7.1 Identified Nouns:

| Serial | Noun             |
|--------|------------------|
| 1      | User             |
| 2      | Authentication   |
| 3      | Profile          |
| 4      | Name             |
| 5      | Email            |
| 6      | Password         |
| 7      | System           |
| 8      | Credential       |
| 9      | Database         |
| 10     | Dashboard        |
| 11     | Account          |
| 12     | Account Settings |

|    |                         |
|----|-------------------------|
| 13 | Session                 |
| 14 | Detailed information    |
| 15 | Logout                  |
| 16 | Images of certificate   |
| 17 | Images of grades        |
| 18 | Images of resume        |
| 19 | Images of CV            |
| 20 | Option ( Hide details ) |
| 21 | Information             |
| 22 | Student                 |
| 23 | Study group             |
| 24 | Group name              |
| 25 | Subject or topic focus  |
| 26 | Group status            |
| 27 | User id                 |
| 28 | Invitation              |
| 29 | Search result           |
| 30 | Search panel            |
| 31 | Notification            |

|    |                    |
|----|--------------------|
| 32 | Group admin        |
| 33 | Member             |
| 34 | Group details      |
| 35 | Description        |
| 36 | Communication      |
| 37 | Chat-system        |
| 38 | Text message       |
| 39 | URL                |
| 40 | Resource-link      |
| 41 | Discussion topic   |
| 42 | Chat-panel         |
| 43 | Study materials    |
| 44 | Files              |
| 45 | Central repository |
| 46 | Discussion thread  |
| 47 | Group panel        |
| 48 | Video call         |
| 49 | Audio call         |
| 50 | Recommendation     |

|    |                           |
|----|---------------------------|
| 51 | Group search panel        |
| 52 | Topics (filter)           |
| 53 | Size (filter)             |
| 54 | Majority members (filter) |
| 55 | Content                   |
| 56 | Group join                |
| 57 | Group join request        |
| 58 | File type                 |
| 59 | File size                 |
| 60 | File name                 |
| 61 | File description          |
| 62 | Upload time               |
| 63 | Role                      |

## 7.2 Identified Verbs:

| Serial | Verbs    |
|--------|----------|
| 1      | create   |
| 2      | provide  |
| 3      | store    |
| 4      | complete |
| 5      | log      |

|    |              |
|----|--------------|
| 6  | authenticate |
| 7  | forget       |
| 8  | use          |
| 9  | guide        |
| 10 | reset        |
| 11 | change       |
| 12 | end          |
| 13 | view         |
| 14 | add          |
| 15 | hide         |
| 16 | ensure       |
| 17 | specify      |
| 18 | select       |
| 19 | assume       |
| 20 | sent         |
| 21 | has          |
| 22 | make         |
| 23 | mean         |
| 24 | join         |
| 25 | search       |
| 26 | accept       |
| 27 | become       |
| 28 | assign       |
| 29 | help         |
| 30 | have         |
| 31 | remove       |
| 32 | update       |
| 33 | interact     |
| 34 | allow        |
| 35 | support      |

|    |          |
|----|----------|
| 36 | share    |
| 37 | post     |
| 38 | discuss  |
| 39 | need     |
| 40 | close    |
| 41 | visible  |
| 42 | place    |
| 43 | upload   |
| 44 | download |
| 45 | delete   |
| 46 | contain  |
| 47 | receive  |
| 48 | join     |
| 49 | start    |
| 50 | reject   |
| 51 | initiate |
| 52 | supports |
| 53 | allow    |
| 54 | invite   |
| 55 | discover |
| 56 | filter   |

### 7.3 General Classification:

Candidate classes are categorized based on the seven general classification. The analysis classes manifest themselves in one of the following ways:

1. External entities
2. Things
3. Events
4. Roles

5. Organizational units

6. Places

7. Structures

A candidate class is selected for special classification if it fulfills two or more characteristics.

| Serial | Solution Space Nouns   | General Classifications |
|--------|------------------------|-------------------------|
| 1      | User                   | 4,5,7(selected)         |
| 2      | Authentication         | 3                       |
| 3      | Profile                | 2,7(selected)           |
| 4      | Name                   | 2                       |
| 5      | Email                  | 2                       |
| 6      | Password               | 2                       |
| 7      | Session                | 2                       |
| 8      | Detailed information   | 2                       |
| 9      | Images of grades       | 2                       |
| 10     | Images of resume       | 2                       |
| 11     | Study Group            | 2,5,7(selected)         |
| 12     | Group name             | 2                       |
| 13     | Subject or topic focus | 2                       |

|    |                    |                 |
|----|--------------------|-----------------|
| 14 | Group status       | 2               |
| 15 | User id            | 2               |
| 16 | Invitation         | 3               |
| 17 | Notification       | 3,7(selected)   |
| 18 | Member             | 4,5,7(selected) |
| 19 | Description        | 2               |
| 20 | Communication      | 3               |
| 21 | Chat               | 2,3,7(selected) |
| 22 | Text message       | 2               |
| 23 | URL                | 2               |
| 24 | Resource-link      | 2               |
| 25 | Discussion topic   | 2               |
| 26 | Chat-panel         | 2               |
| 27 | Files              | 2,7(selected)   |
| 28 | Central repository | 2,7(selected)   |
| 29 | Group panel        | 2               |
| 30 | Video call         | 3               |
| 31 | Audio call         | 3               |

|    |                           |   |
|----|---------------------------|---|
| 32 | Recommendation            | 2 |
| 33 | Group search              | 3 |
| 34 | Topics (filter)           | 2 |
| 35 | Size (filter)             | 2 |
| 36 | Majority members (filter) | 2 |
| 37 | Content                   | 2 |
| 38 | Group join                | 3 |
| 39 | Group join request        | 3 |
| 40 | File type                 | 2 |
| 41 | File size                 | 2 |
| 42 | File name                 | 2 |
| 43 | File description          | 2 |
| 44 | Upload time               | 2 |
| 45 | Role                      |   |

## 7.4 Selection Criteria:

The candidate classes are then selected as classes by six Selection Criteria. A candidate class generally becomes a class when it fulfills around three characteristics.

1. Retain information
2. Needed services
3. Multiple attributes

4. Common attributes
5. Common operations
6. Essential requirements

Potential general classified nouns to become a class after selection criteria :

| <b>Serial</b> | <b>Solution Space Nouns</b> | <b>Selection Criteria</b> |
|---------------|-----------------------------|---------------------------|
| 1             | <b>User</b>                 | 1-5(selected)             |
| 2             | Profile/ <b>UserProfile</b> | 1-5(selected)             |
| 3             | Group                       | 1-5(selected)             |
| 4             | GroupMember                 | 1-5(selected)             |
| 5             | Chat/GroupChat              | 1-5(selected)             |
| 6             | Notification                | 1-5(selected)             |
| 7             | File*                       | 1-5(selected)             |
| 8             | FileRepository*             | 1-5(selected)             |

Additional classes required :

| <b>Serial</b> | <b>Additional classes</b> |
|---------------|---------------------------|
| 1             | UserService               |
| 2             | SessionService            |
| 3             | PasswordService           |
| 4             | AuthService               |
| 5             | UserProfileService        |
| 6             | GroupRecommendation*      |
| 7             | GroupSearch               |
| 8             | VideoCall*                |

## 7.5 Class Card:

| <b>User</b>                                                                                                    |                                                                                |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Attribute</b>                                                                                               | <b>Method</b>                                                                  |
| -Id<br>-Name<br>-Email                                                                                         | +getId()<br>+setId()<br>+getName()<br>+setName()<br>+getEmail()<br>+setEmail() |
| <b>Responsibilities</b>                                                                                        | <b>Collaborator</b>                                                            |
| <ul style="list-style-type: none"> <li>• Acts as a unit of user</li> <li>• Contains basic user info</li> </ul> | Student, Teacher                                                               |

| <b>UserService</b> |               |
|--------------------|---------------|
| <b>Attribute</b>   | <b>Method</b> |

|                                                                                                                             |                                                |
|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
|                                                                                                                             | +addUser()<br>+removeUser()<br>+retrieveUser() |
| <b>Responsibilities</b>                                                                                                     | <b>Collaborator</b>                            |
| <ul style="list-style-type: none"> <li>• Add or remove user from database</li> <li>• Retrieve user form database</li> </ul> |                                                |

| <b>SessionService</b>                                                                 |                                                                                                                                              |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attribute</b>                                                                      | <b>Method</b>                                                                                                                                |
|                                                                                       | +generateSessionToken()<br>+retrieveSessionToken()<br>+addSessionToken()<br>+validateSessionToken<br>+invalidateSessionToken<br>+isExpired() |
| <b>Responsibilities</b>                                                               | <b>Collaborator</b>                                                                                                                          |
| <ul style="list-style-type: none"> <li>• Manages session related functions</li> </ul> |                                                                                                                                              |

| <b>PasswordService</b> |               |
|------------------------|---------------|
| <b>Attribute</b>       | <b>Method</b> |

|                                                                                      |                                                                                                                                                                                |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -password                                                                            | +checkPassword()<br>+isValidPassword()<br>+retrieveHashedPassword()<br>+hashPassword()<br>+changePassword()<br>+resetPassword()<br>+generateOTP()<br>+sendResetPasswordEmail() |
| <b>Responsibilities</b>                                                              | <b>Collaborator</b>                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>Manages password related functions</li> </ul> |                                                                                                                                                                                |

| <b>AuthService</b> |                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------|
| <b>Attribute</b>   | <b>Method</b>                                                                                        |
|                    | +register()<br>+login()<br>+isAuthenticated()<br>+logout()<br>+updatePassword()<br>+forgotPassword() |

| <b>Responsibilities</b>                                                                                                | <b>Collaborator</b> |
|------------------------------------------------------------------------------------------------------------------------|---------------------|
| <ul style="list-style-type: none"> <li>• Login and register user</li> <li>• Manages forgot password process</li> </ul> | User                |

| <b>UserProfile</b>                                                                                                                                                                           |                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attribute</b>                                                                                                                                                                             | <b>Method</b>                                                                                                                                                             |
| -userId<br>-date_of_birth<br>-age<br>-gender<br>-phone_number<br>-profile_pictureURL<br>-gradeSheetURL<br>-resumeURL<br>-address<br>-bio<br>-department<br>-education<br>-role<br>-isVisible | setter and getter for all field<br>+addProfilePicture()<br>+getProfilePicture()<br>+addGradeSheet()<br>+getGradeSheet()<br>+addResume()<br>+getResume()<br>+hideProfile() |
| <b>Responsibilities</b>                                                                                                                                                                      | <b>Collaborator</b>                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>• Contains detail info of user</li> </ul>                                                                                                             | User                                                                                                                                                                      |

| <b>UserProfileService</b> |               |
|---------------------------|---------------|
| <b>Attribute</b>          | <b>Method</b> |

|                                                                                      |                                                                                                      |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|                                                                                      | +doesUserExist()<br>+doesProfileExist()<br>+createProfile()<br>+updateProfile()<br>+getProfileInfo() |
| <b>Responsibilities</b>                                                              | <b>Collaborator</b>                                                                                  |
| <ul style="list-style-type: none"> <li>Create, update or get profile info</li> </ul> | User                                                                                                 |

| <b>Group</b>                                                                                                      |                                                                            |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Attribute</b>                                                                                                  | <b>Method</b>                                                              |
| -group_name<br>-group_description<br>-group_status<br>-type<br>-topics<br>-group_image<br>-created_by             | +createGroup()<br>+retrieveGroupInfo()<br>+updateGroup()<br>+removeGroup() |
| <b>Responsibilities</b>                                                                                           | <b>Collaborator</b>                                                        |
| <ul style="list-style-type: none"> <li>Manages group related functions</li> <li>Create or remove group</li> </ul> | User                                                                       |

|                                                                            |  |
|----------------------------------------------------------------------------|--|
| <ul style="list-style-type: none"> <li>Retrieve or update group</li> </ul> |  |
|----------------------------------------------------------------------------|--|

| <b>GroupMembers</b>                                                                                                                                          |                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attribute</b>                                                                                                                                             | <b>Method</b>                                                                                                                                                                                    |
| -groupId<br>-members[]<br>-admins[]                                                                                                                          | +addAdmin()<br>+joinMember()<br>+removeMember()<br>+retrieveAllGroupMember()<br>+retrieveAllAdmins()<br>+getAllGroupsOfMember()<br>+updateMemberRole()<br>+getUserRole()<br>+_getMembersByRole() |
| <b>Responsibilities</b>                                                                                                                                      | <b>Collaborator</b>                                                                                                                                                                              |
| <ul style="list-style-type: none"> <li>Contains all group members</li> <li>Join or remove members in group</li> <li>Add or remove admins in group</li> </ul> | User, GroupAdmin                                                                                                                                                                                 |

| <b>GroupChat</b> |               |
|------------------|---------------|
| <b>Attribute</b> | <b>Method</b> |

|                                                                                                                                                                 |                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -groupId<br>-chat[] (Chat)                                                                                                                                      | +postMessage()<br>+removeChat()<br>+fileUpload()<br>+postDiscussion()<br>+retrieveAllChat()<br>+retrieveLatestChat()<br>+retrieveAllDiscussion()<br>+checkDiscussionExists()<br>+retrieveChats() |
| <b>Responsibilities</b>                                                                                                                                         | <b>Collaborator</b>                                                                                                                                                                              |
| <ul style="list-style-type: none"> <li>• Contains all group chats</li> <li>• PostChat or discussion by users</li> <li>• Retrieve chats from database</li> </ul> |                                                                                                                                                                                                  |

| <b>Notification</b>                                 |                                               |
|-----------------------------------------------------|-----------------------------------------------|
| <b>Attribute</b>                                    | <b>Method</b>                                 |
| -type<br>-content<br>-sender(User)<br>-receive_date | +storeNotification()<br>+deleteNotification() |

| <b>Responsibilities</b>                                                                                                                                                   | <b>Collaborator</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <ul style="list-style-type: none"> <li>• Acts as a single notification entity</li> <li>• Contains notification info</li> <li>• SendNotification for all events</li> </ul> |                     |

| <b>File</b>                                                                      |                                  |
|----------------------------------------------------------------------------------|----------------------------------|
| <b>Attribute</b>                                                                 | <b>Method</b>                    |
| -name<br>-type<br>-size<br>-description<br>-pathURL                              | getter and setter for all fields |
| <b>Responsibilities</b>                                                          | <b>Collaborator</b>              |
| <ul style="list-style-type: none"> <li>• Acts as a single file entity</li> </ul> |                                  |

| <b>FileRepository</b> |               |
|-----------------------|---------------|
| <b>Attribute</b>      | <b>Method</b> |

|                                                                                                                                                     |                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| -user_id                                                                                                                                            | +retrieveAllUserFile()<br>+retrieveFile() |
| <b>Responsibilities</b>                                                                                                                             | <b>Collaborator</b>                       |
| <ul style="list-style-type: none"> <li>• Contains all the saved files</li> <li>• Organize the files</li> <li>• Download files to machine</li> </ul> |                                           |

| <b>VideoCall</b>                                                                                                                            |                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>Attribute</b>                                                                                                                            | <b>Method</b>                                  |
|                                                                                                                                             | +setUpWebRTC()<br>+initateCall()<br>+endCall() |
| <b>Responsibilities</b>                                                                                                                     | <b>Collaborator</b>                            |
| <ul style="list-style-type: none"> <li>• Manage video call</li> <li>• Setup external technology</li> <li>• Initiate and end call</li> </ul> |                                                |

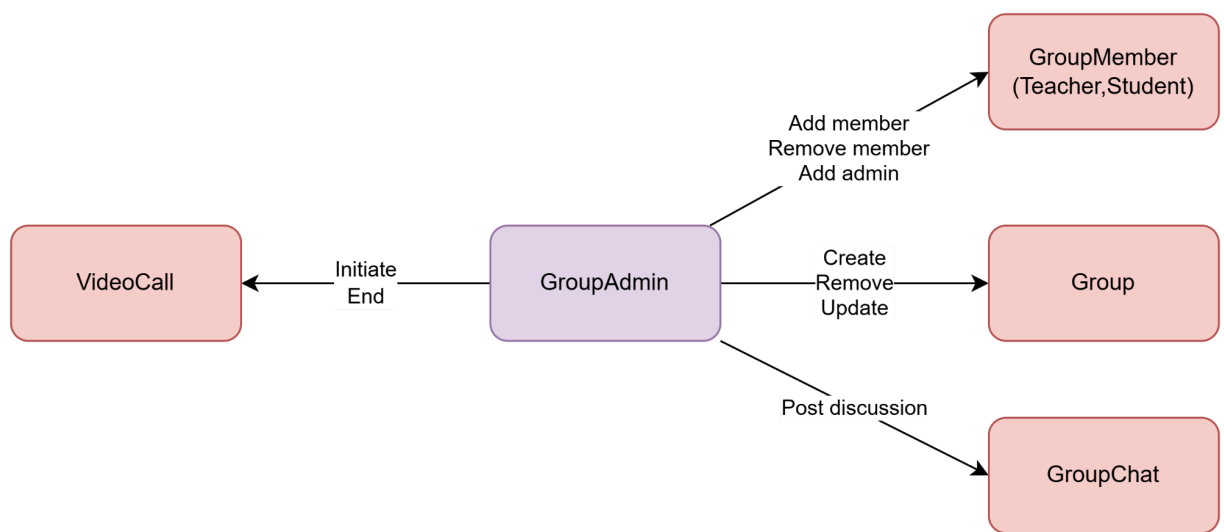
| <b>GroupRecommendation</b>                                                                                                       |                                                  |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Attribute</b>                                                                                                                 | <b>Method</b>                                    |
| -groupType                                                                                                                       | +retrieveAllGroups()<br>+findRecommendedGroups() |
| <b>Responsibilities</b>                                                                                                          | <b>Collaborator</b>                              |
| <ul style="list-style-type: none"> <li>• Manage user data for recommendation</li> <li>• Retrieve groups from database</li> </ul> |                                                  |

| <b>GroupSearch</b>                                                          |                                        |
|-----------------------------------------------------------------------------|----------------------------------------|
| <b>Attribute</b>                                                            | <b>Method</b>                          |
| -type<br>-topics<br>-size                                                   | +searchGroup()<br>+retrieveAllGroups() |
| <b>Responsibilities</b>                                                     | <b>Collaborator</b>                    |
| <ul style="list-style-type: none"> <li>• Filter groups by search</li> </ul> |                                        |

## 7.8 CRC Diagram:

Diagram Id: 1

Name: GroupAdmin



**Fig 21:** CRC(GroupAdmin)

Diagram Id: 2

Name: Teacher,Student(GroupMember)

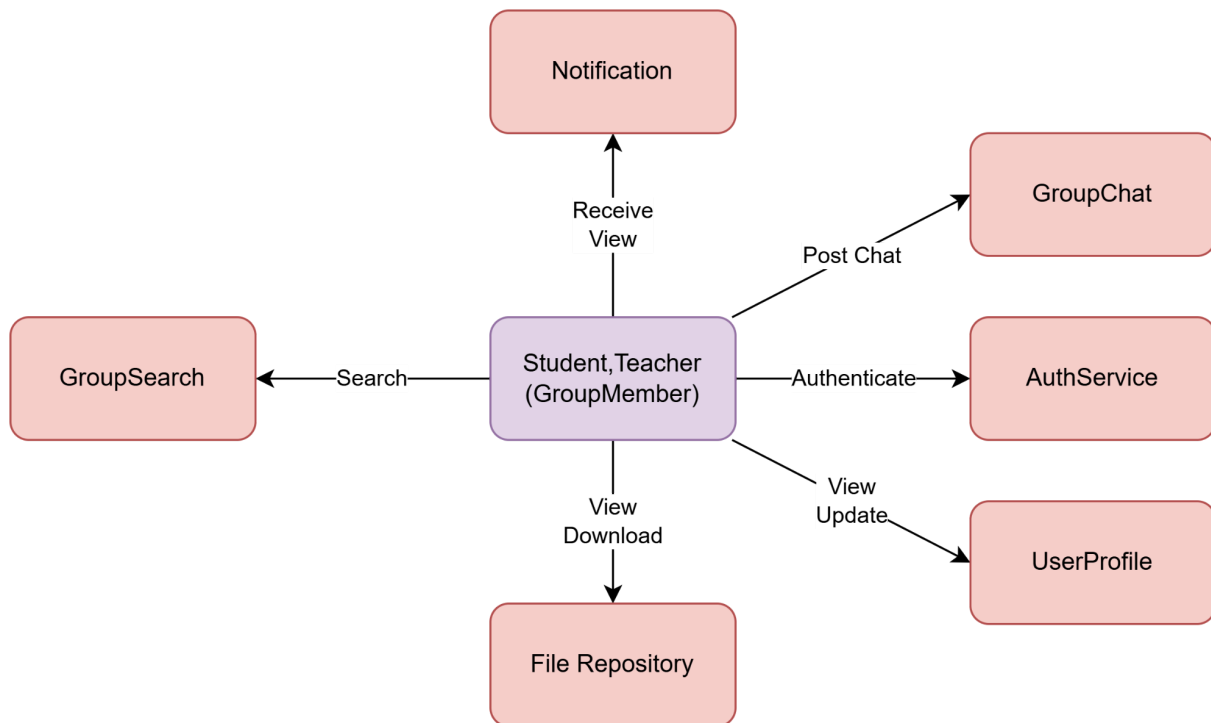


Fig 22: CRC(Teacher,Student)

## 8. Behavioral Diagram

### 8.1 Event Table :

| Serial | Event | Initiator | Associated |
|--------|-------|-----------|------------|
|--------|-------|-----------|------------|

|    |                 |             | <b>Methods</b>                                 |
|----|-----------------|-------------|------------------------------------------------|
| 1  | Login           | User        | +register()                                    |
| 2  | Register        | User        |                                                |
| 3  | Join Group      | User        | +createGroup()                                 |
| 4  | Create Group    | Group Admin |                                                |
| 5  | Update Group    | Group Admin | +createGroup()<br>+joinGroup()                 |
| 6  | Remove Group    | Group Admin | +createGroup()                                 |
| 7  | Add Admin       | Group Admin | +createGroup()                                 |
| 8  | Add member      | Group Admin | +register()<br>+createGroup()                  |
| 9  | Remove member   | Group Admin | +createGroup()<br>+addMember()                 |
| 10 | Update Profile  | User        | +createProfile()                               |
| 11 | Post Chat       | User        | +createGroup()<br>+sendNotiifacation           |
| 12 | Post Discussion | Group Admin | +createGroup()<br>+sendNotification()          |
| 13 | Upload File     | User        | +sendNotification()                            |
| 14 | Download File   | User        | +uploadFile()                                  |
| 15 | Search Group    | User        | +createGroup()                                 |
| 16 | Video Call      | Group Admin | +setUpWebRTC()<br>+initateCall()<br>+endCall() |

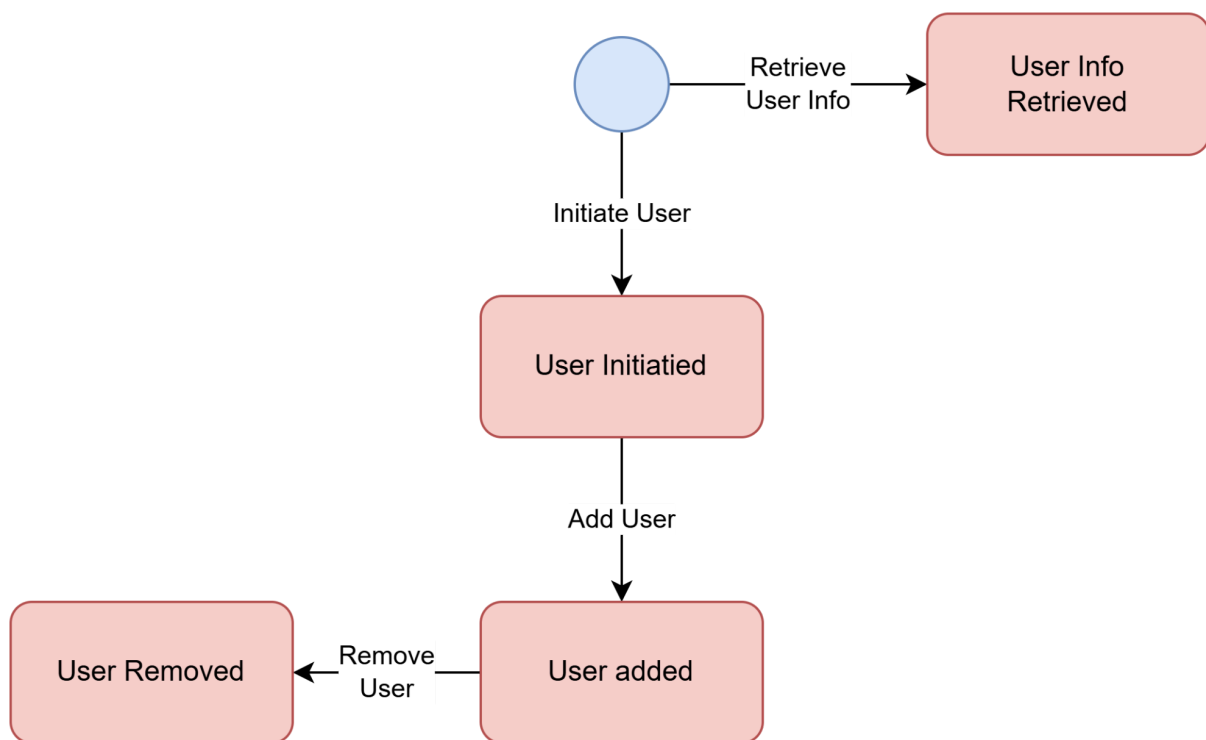
## 8.2 State Transition Diagram

A UML state diagram, which depicts the active states for each class and the occasions (triggers) that induce changes between these active states, is part of a behavioral

model. Here we've listed the most important events that trigger the change of states.

### Diagram ID: 1

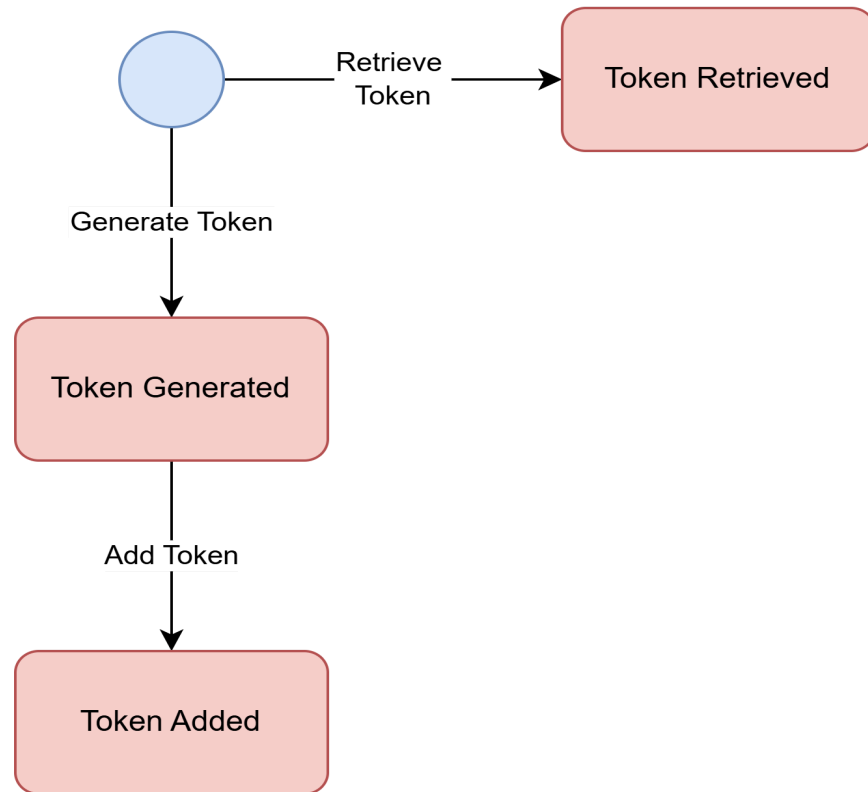
**Class Name: UserService**



**Fig 23: UserService State Transition**

### Diagram ID: 2

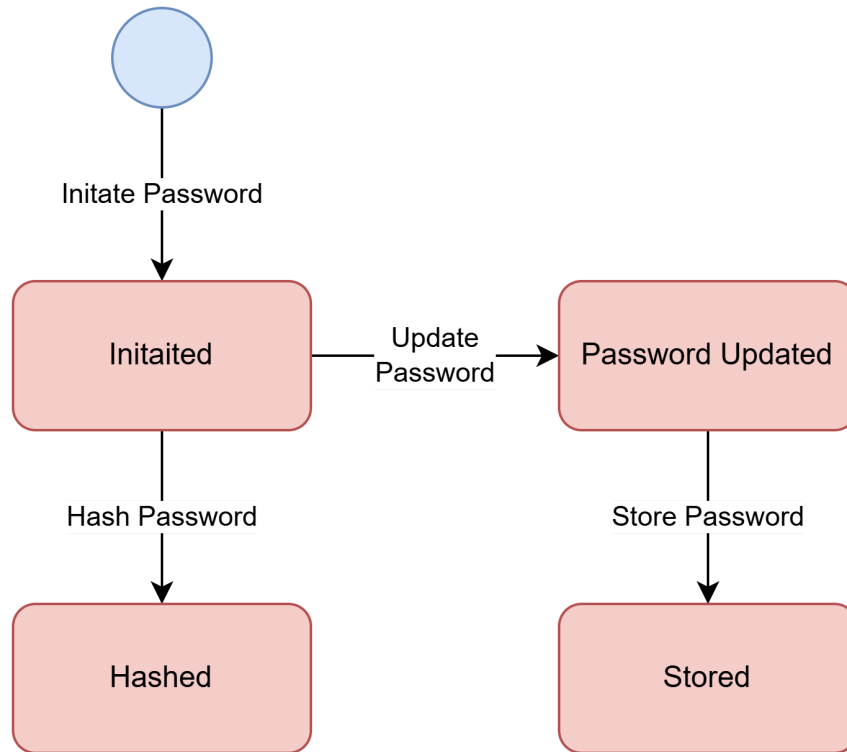
**Class Name: SessionService**



**Fig 24: SessionService** State Transition

**Diagram ID: 3**

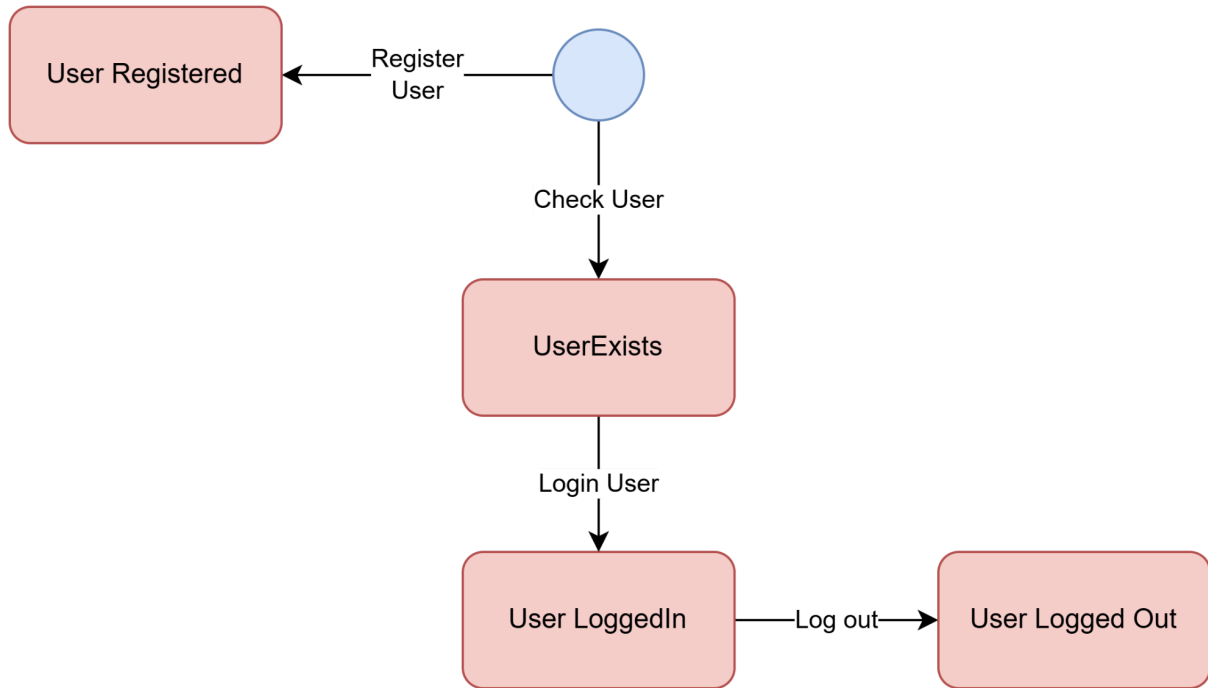
**Class Name: PasswordService**



**Fig 25: PasswordService State Transition**

**Diagram ID: 4**

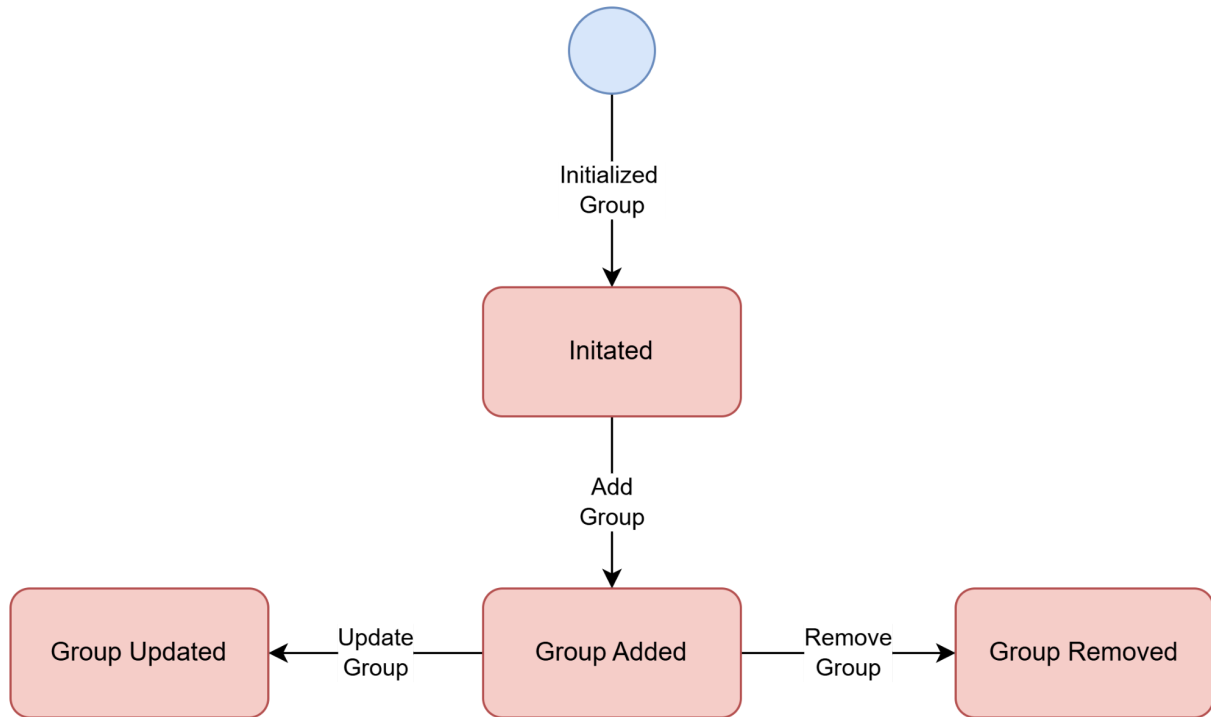
**Class Name: AuthService**



**Fig 26: AuthService State Transition**

**Diagram ID: 5**

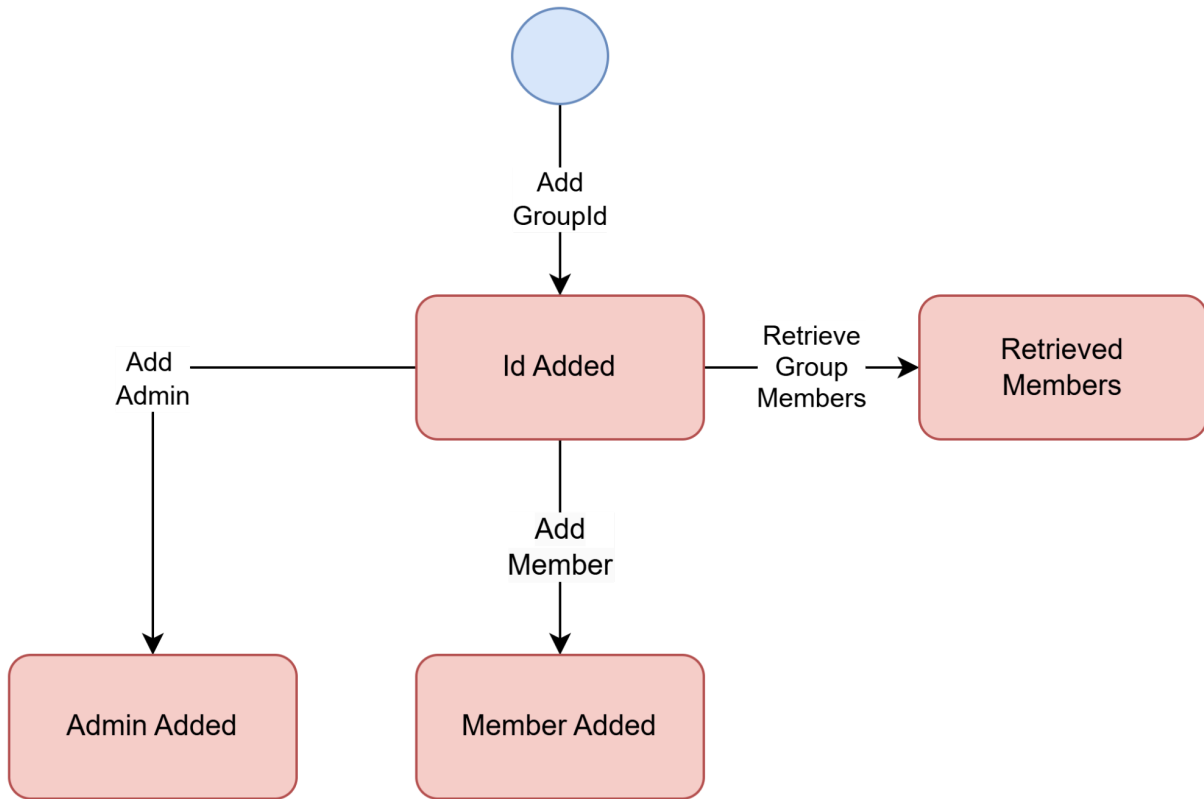
**Class Name: Group**



**Fig 27: Group State Transition**

**Diagram ID: 6**

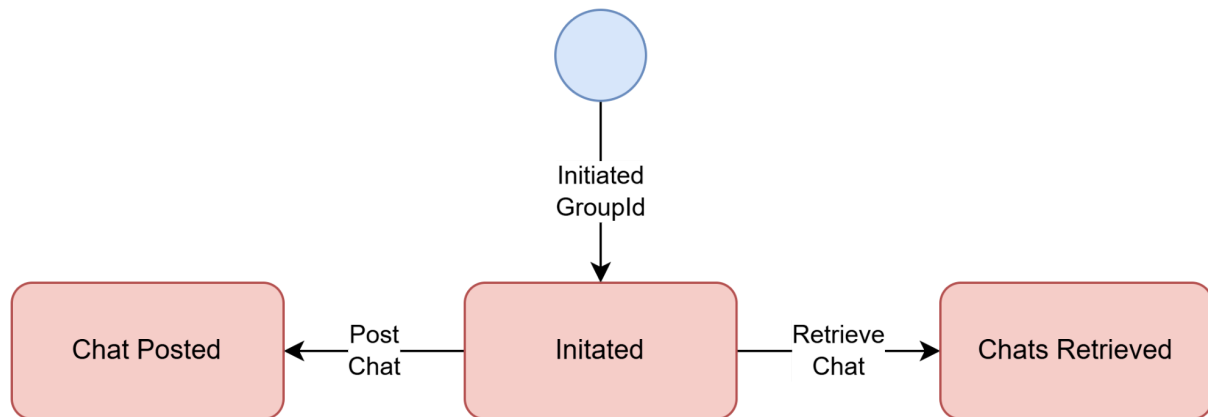
**Class Name: GroupMembers**



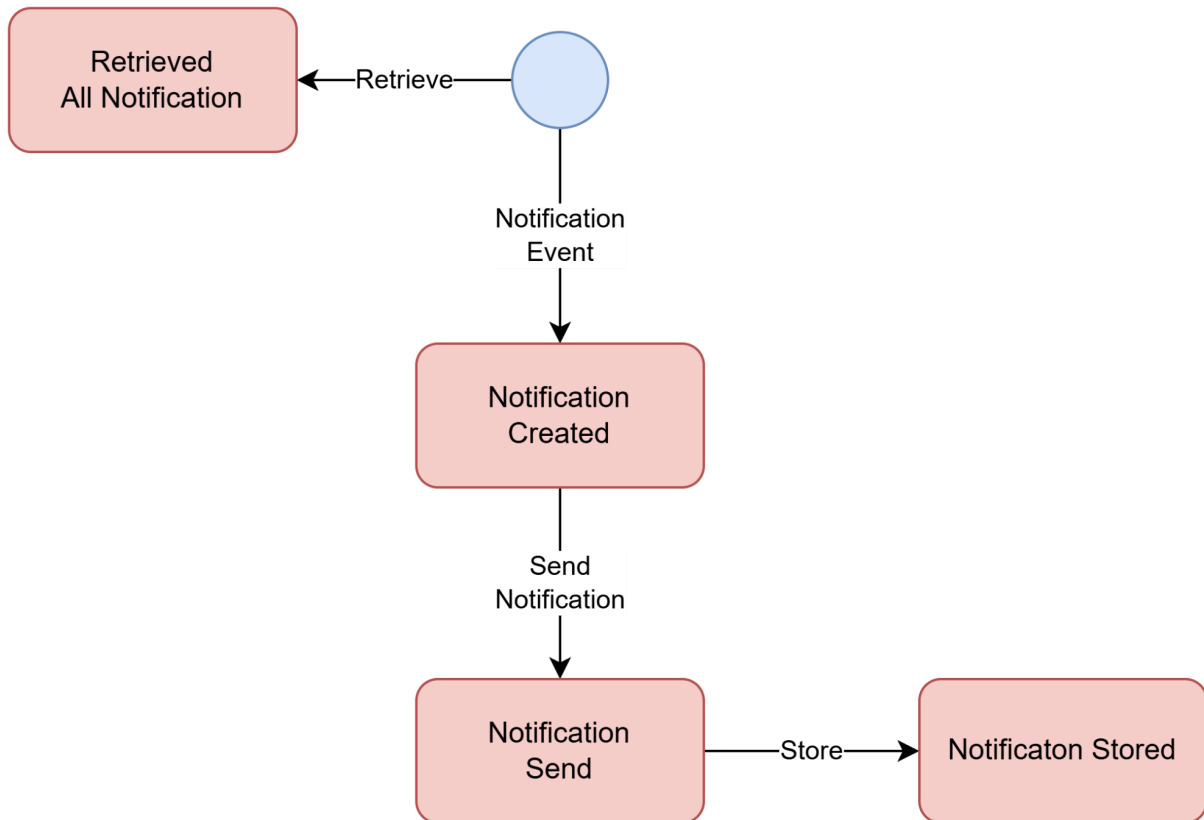
**Fig 28: GroupMember State Transition**

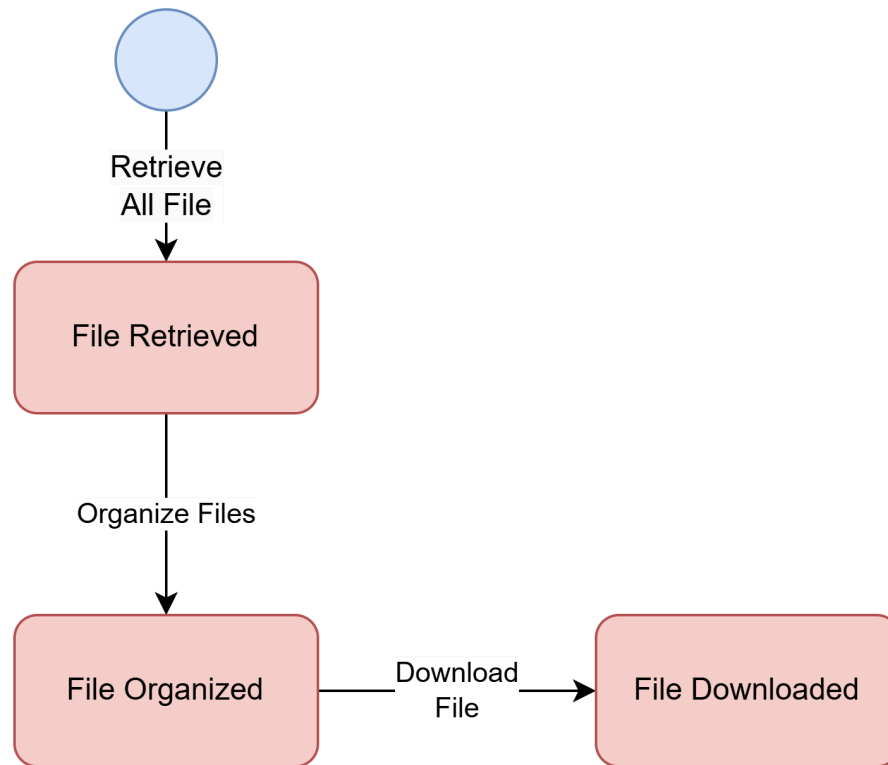
**Diagram ID: 7**

**Class Name: GroupChat**



**Fig 29: GroupChat State Transition**

**Diagram ID: 8****Class Name: Notification****Fig 30: Notification State Transition****Diagram ID: 9****Class Name: FileRepository**



**Fig 31: FileRepository** State Transition

### 8.3 Sequence Diagram

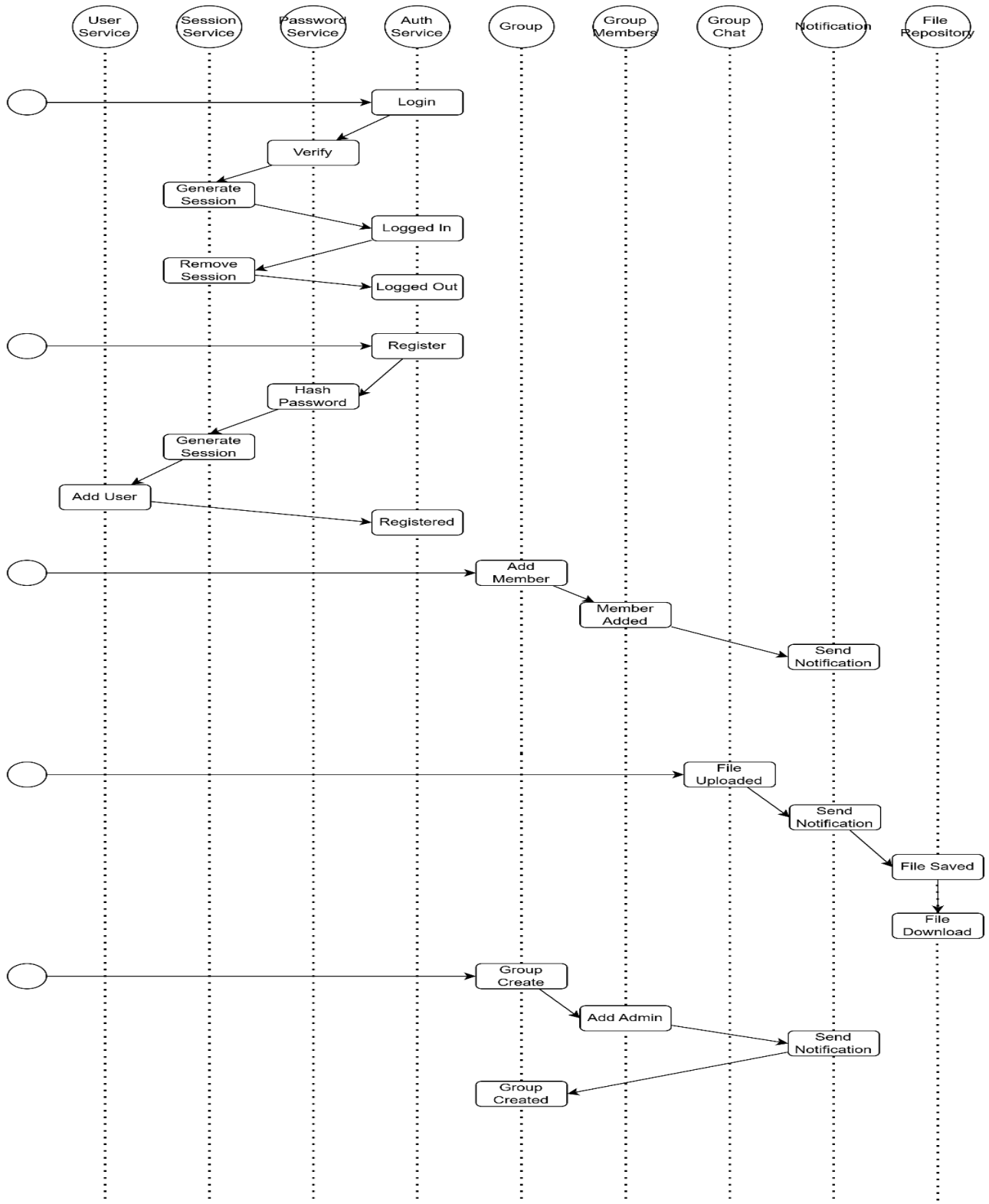
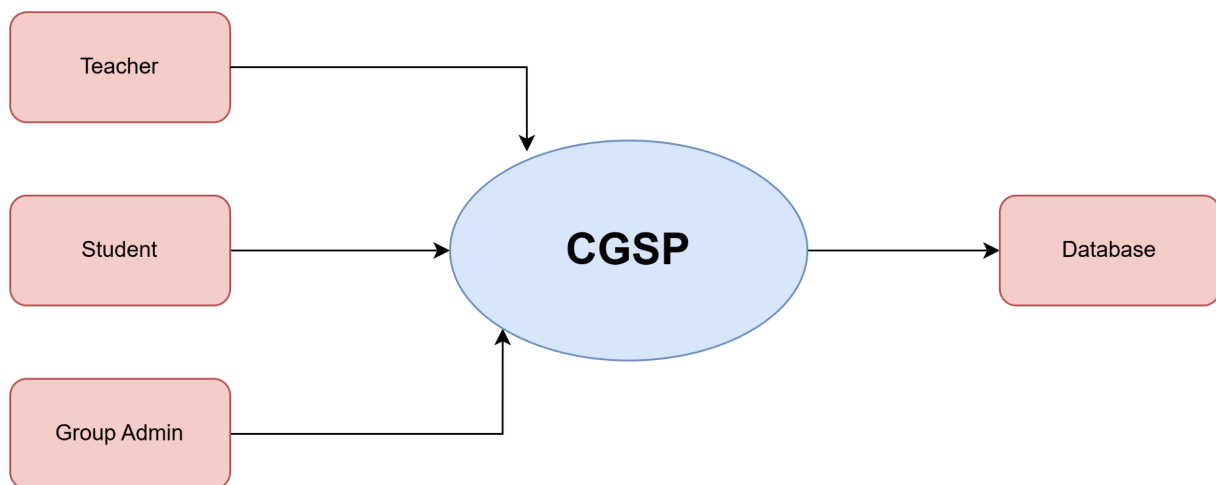


Fig 32: Sequence Diagram

## 9. Data Flow Diagram

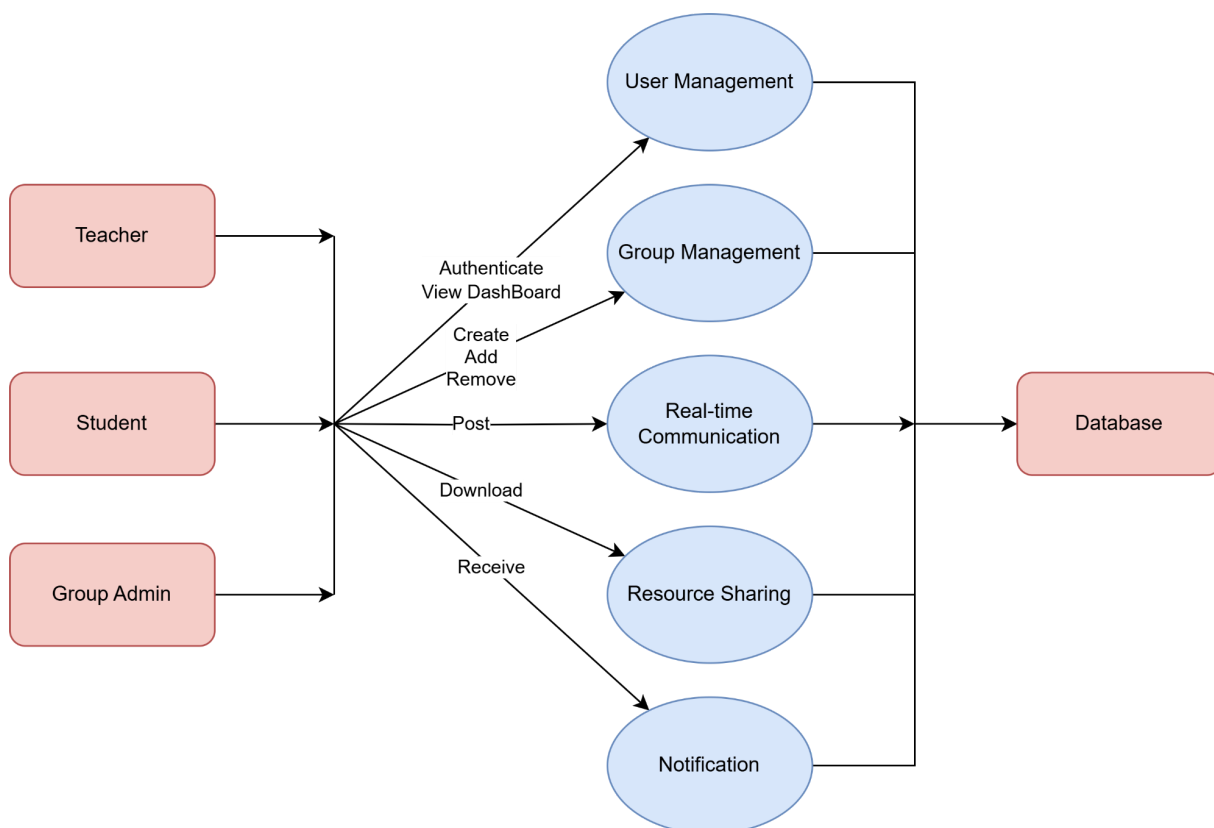
A data-flow diagram is a visual representation of how data moves through a system or a process. A data flow diagram (DFD) shows how information moves through any system or process. It displays data inputs, outputs, storage locations, and routes between each destination using predefined symbols such as rectangles, circles, and arrows as well as brief text labels.

### 9.1 Level 0: Collaborative Group Study Platform



**Fig 33:** Data Flow Diagram

### 9.2 Level 1: Collaborative Group Study Platform



**Fig 34:** Data Flow Diagram